

Understanding Futures: A deeper insight into the current dynamics of futures & strategy design

By: Isaac

Overview

Currently the retail space within futures is messy. There are numerous individuals claiming a specific ruleset or strategy is the key to winning, there are those following concepts or market dynamics that are mostly pseudoscience or just older strategies with a new label slapped on them. There are con-men everywhere you look. Most traders give up, with current statistics placing the actual pool of profitable traders at sub 1%. The entirety of the market uses retail traders as fuel, meanwhile TJR sells hopes and dreams to those mindless enough to believe him. Today I offer you this, an insight into the actual data, the math that actually drives this data and its corresponding movement of price and how you can use it to find your own edge, along with where you should begin with all of it.

This paper will be structured simply. We will begin with debunking what is worth your time and what isn't, then proceed by diving layer by layer into how to properly utilize what is actually worth your time, how to understand it, and even how to build a strategy on it.

Before we begin, I want to ask you to do one thing. Detach from the idea of a strategy as a singular set or basis/criteria that is concrete. Concrete strategies are equivalent to monetary obliteration in the current retail space unless it's simply buy and hold (but that doesn't apply to futures- so I'll move on).

Debunking current fallacies

1. ICT Concepts:

A trading methodology designed by Michael J. Huddleston. Provides beginners with a more "understandable" and "simple" way to explain larger market concepts & theory but the framework often suffers from rebranded century-old ideas. After entering into multiple trading contests (and subsequently failing all of them), he proceeds to preach inconsistent trading methods leaving students unprofitable, confused, or simply blindly believing eventually the right setup will come.

Numerous ICT "terms" have simply been re-branded from already known concepts. Remember, these concepts existed but as *concepts* not as a basis for a strategy. The official definition of a concept is: "an abstract idea; a general notion". ICT concepts re-brand ideas and abstract thoughts as strategy and assumes they will hold up. ICT or "michael" claims that a secret banking algorithm called IPDA controls the entire global market specifically to trap retail traders, a baseless claim that has zero evidence and isn't even profitable for banks if this were the case.

Ultimately you would think that someone like ICT would attempt to back up his claims. Which he did by claiming to show his verified broker statement which was later to be proved as photoshopped along with multiple trading screenshots found to directly be taken from a simulated account instead of a real trading panel. After failed challenges, a failed 2016 and 2024 robbins cup, it is without a doubt this is not the guy you should be betting is going to give you a winning strategy.

2. The “Smart Money” Conspiracy Ecosystem

Beyond general ICT narrative's, there is an entire genre of “smart money” mythology. Many traders utilize these as either coping mechanisms or concepts to derive reason as to why their gamble-of-a-trade went well. Often terms like the following are thrown around:

“Whales are hunting your stops” - The belief that market makers specifically or remotely give half a shit about retail stop-loss clusters is hilarious. In reality, stop hunts are a natural consequence of liquidity dynamics, not a coordinated vendetta against retail traders.

“Insider trading drives everything” - This is a half-truth but an incredibly hard one to prove. The idea that every price move is traceable to a hidden institutional agenda is unfalsifiable and conspiratorial. This goes along with 1930s era concepts of a single manipulative entity controlling the market. Taught by many gurus as some identifiable force, but overall remaining an idea rather than an actual explanation as to “why” something happened.

3. Gann Theory & Elliott wave Theory

W.D. Gann's methods, composing mainly of angles, squares, harmonics, planetary circles, and numerology applied to price charts. It primarily remains popular as a smaller lesser known perspective within trading. Gann claims %90 accuracy but none of this has been independently verified. His methods remain vague and multi-layered enough so that every market move can be somehow magically “magically” explained by some Gann technique. This isn't even going into how the astrological components of Gann analysis are quite literally pseudoscience.

Elliott wave counting is so subjective that multiple analysts can arrive at completely different wave counts on the same chart, with both claiming to be “correct” after the fact. There is zero statistical evidence that Elliott wave patterns have predictive power beyond random chance. The theory itself is complex and serves as a protective mechanism for when predictions fail, allowing practitioners to blame an incorrect wave count rather than theory itself.

4. Fibonacci levels & Harmonics

Fibonacci retracement levels (38.2, 50, 61.8) appear to work largely because enough traders watch them, allowing them to be a self-fulfilling prophecy, not a mathematical law of markets. Studies have shown that randomly generated levels produce similar “bounce” rates as Fibonacci levels, suggesting it is more perceived significance coupled with confirmation bias doing the real work here.

Harmonic patterns require precise Fibonacci ratios to be “valid” but their subjective identification makes them unbacktestable in practice. When a harmonic pattern “fails,” harmonic practitioners simply re-classify it as a different pattern, once again another unfalsifiable framework.

5. Win Rate culture & Prop firm Illegitimacy

Prop firm marketing & Social media have propagated the idea that high win rates are necessary or even desirable. For example: A 40% win rate with 3:1 reward-to-risk is mathematically superior to a %90 win rate with inverted risk, yet the retail community fetishizes

win percentage. This myth is perpetuated by screenshot culture, traders post winning streaks (hot-hand fallacy) while hiding their equity curves.

The modern prop firm model (FMO, Apex, Topstep, etc) profits primarily from evaluation fees, not from profitable traders that passed evals. The business model is closer to a casino than an actual traditional prop desk. Passing a challenge is often marketed as a proof of trading skill, and while it is to a certain extent, evaluation parameters frequently encourage reckless high leverage gambling behavior. Many “funded traders” are immediately reset back to evaluation status allowing the prop firm to create a revolving door of fees. Not to mention many traders that do become consistently profitable, get kicked off of prop firms for random rule violations that do not exist.

6. Algos, Cognitive Fallacies & False Adages

Countless retail “algo” products are marketed with backtested equity curves that are purely designed for curve-fitting, optimized to past data with zero forward validity. If a bot genuinely produced consistent alpha, its creators would have zero rational reason to sell it for 99\$ a month.

These aren’t strategies per se, but they are systematically propagated by the retail trading ecosystem and often end up being the downfall of most traders, starting with the Gamblers Fallacy: “Price has dropped 5 days in a row now, so it must bounce”. The Hot-Hand Fallacy: “I’m on a winning streak so I should increase my contract sizing.” The Sunk-Cost Fallacy: “I’ve invested too much into this position/education, I can’t quit now.” Survivorship bias is a big one with social media showing only the winners, often distorting the perceptions of how likely success actually is and glorifying those who gamble and make big wins. Lastly, confirmation bias, basically traders seeking out analysis that agrees with their existing position and dismissed contradictory evidence.

The Uncomfortable Truth

The dirty and uncomfortable truth is that the vast majority of retail futures traders lose money over time. Regulatory disclosures from brokers consistently show that 70-90% of retail accounts trading leveraged derivatives lose money. The entire ecosystem of gurus, signal sellers, prop firms, discord groups, and indicator vendors exist to monetize the dream of trading success, not for deliverance. The common thread across every fallacy above is they provide narrative comfort as a coping mechanism for the lack of understanding of a chaotic system without statistical evidence of an edge.

The truth is, a trading claim should be treated as unproven unless it specifies its dynamics in advance, accounts for fees and slippage, and survives out-of-sample testing across market regimes.

Defining what we know

Naturally our reaction to this is somewhat one of discouragement. “So if most everything I’ve been shown by youtube, other traders, all of it is a lie, what is true?” To this I give you 2 routes. There are two routes that are real. And it is important to understand the difference between the two so you can pick which side you choose to embark on your journey to profit with. The first being discretionary traders. There are many traders who are profitable and claim it's because of some setup or strategy, the simple and hard to believe truth for them is NOT that they have discovered some profitable strategy, in fact it probably has little to nothing to do with their core strategy, it has to do with the mental training & pattern

recognition. Not chart patterns themselves, but the actual neurons and synapses firing off at a neurological level. The training of your mind. Often this is done through pure market exposure, years of time, or some kind of genius level thinking. For most though this isn't achieved through genius, it's achieved through experience, hard work, and time. Their strategy when taught may not even work for others. It may even fail in backtests, confirming that it's less about the strategy, more about the mental training to actually achieve results. Consistency doesn't look like all green days, it has red days. But ultimately these individuals have trained themselves mentally to observe a market and be profitable. Harder to explain or teach, easier to understand once you get there.

The second reality is the most concrete. This is the mathematics behind the market. This involves layers of Quantitative & Fundamental analysis, analyzing the numbers behind "what made the price move" vs baseless claims. This is often a more desirable route for those with a more logical based perspective, however this is also the deepest and hardest route to go down. This requires a deep will and understanding coupled with an extreme desire to learn. There is a third and final option I believe to be the closest to a "trading holy-grail" you will ever become. It is a combination of quantitative and discretionary routes. The ultimate ability to maintain discretion while having understanding of the market as a set of probabilities and less as a "should I pick up or down".

Today I will be diving into this. The combination of the two, how a synergy can be achieved by combining discretion with a quantitative understanding of the markets, and what we know right now along with how to build on that knowledge.

We will begin by describing levels of data granularity, their advantages & disadvantages, and what you can derive from each level respectively. Data granularity refers to the amount of detail preserved in a dataset, coarse data summarizes many market events into one observation, with fine data recording smaller and more specific events. Within futures markets this ranges from a basic candlestick chart to individual order additions, cancellations, modifications, executions, and queue position. More detail improves observation, but it does not eliminate uncertainty or prove that a recognizable pattern has predictive value, below is a data-granularity ladder:

(located on the next page due to it being large)

Level	What it contains	What you can derive	Advantages	Disadvantages
1. Charting-platform OHLCV	Open, high, low, close, and volume for a selected timeframe; TradingView is the usual retail example	Trend state, range, volatility, prior highs/lows, session high/low, relative volume, basic support/resistance, VWAP and moving-average calculations	Simple, cheap, easy to read, useful for broad context and longer-horizon research	Each candle hides the sequence of trades within it; it cannot identify who initiated trades, resting liquidity, order cancellations, or execution quality
2. Smaller timeframe / intrabar OHLCV	The same OHLCV fields, but broken into smaller bars: e.g., five-minute, one-minute, seconds, ticks, range bars, or volume bars	More detailed auction sequence, intraday volatility, session behavior, opening range, time-of-day effects, more precise backtesting than higher-timeframe bars	Still accessible and relatively easy to store and test; shows whether a higher-timeframe move developed smoothly or violently	More bars means more noise and more opportunities to overfit; ordinary bars still do not reveal individual transactions or the order book
3. Tick / time-and-sales data	Each reported trade: price, size, and timestamp; trade direction may be inferred or supplied depending on feed and method	Trade intensity, volume at price, pace of transactions, large prints, realized volume distribution, volume profile, VWAP, short-term volatility, and potential aggressor imbalance	Moves from summarized candles to actual executed transactions; better for execution research and volume-at-price analysis	A print proves execution, not motive; a large trade may be one participant, a spread, liquidation, or several matches; aggressor classification can be imperfect without richer data
4. Bid/ask or quote data	Best bid and best ask, associated displayed size, and quote updates; often called top-of-book or Level 1	Spread, quote changes, immediate displayed liquidity, short-term imbalance at the inside market, and whether trades occur at bid or ask	Useful for understanding actual tradable prices, not just last-trade price; important for realistic stop and fill assumptions	Only shows the inside market; displayed size is not guaranteed to remain, and it does not reveal depth beyond the best bid and ask

5. Market-by-Price depth (MBP / Level 2 DOM)	Aggregated resting quantity and order count at each visible price level; CME MBP disseminates summary quantity and number of orders for up to 10 price levels	Liquidity concentration, visible depth, near-price imbalance, potential execution conditions, pull/stack behavior, and a basic depth-of-market view	Better context for short-term execution than candles or prints alone; shows where visible limit liquidity is currently resting	Aggregates all orders at a price, so you cannot identify individual order size or exact queue position; displayed liquidity can be canceled, refreshed, or misleading; it is descriptive, not a reliable directional signal by itself.
6. Footprint / volume-at-price analytics	Usually a visualization built from tick and bid/ask data: volume traded at each price, bid-versus-ask volume, delta, cumulative delta, imbalances, and point-of-control measures	Where transactions occurred within a bar, aggressive buying/selling estimates, auction completion or rejection hypotheses, local participation intensity, and location-specific volume analysis	Compresses large amounts of trade data into a readable format; useful for reviewing execution, locating heavy trade, and building testable hypotheses	It is an analytical transformation, not a new source of information; “absorption,” “trapped traders,” and delta-divergence interpretations are hypotheses that require objective testing, not facts revealed by the chart
7. Market-by-Order r (MBO / Level 3)	Every visible individual resting order, including anonymous OrderID, size, price, queue priority, additions, cancellations, modifications, and full book depth	Individual order lifecycle, queue position, replenishment behavior, order-book reconstruction, cancellation rates, liquidity persistence, and higher-quality transaction/quote research	Highest practical public granularity for the visible CME central limit order book; MBO provides individual orders across all price levels rather than aggregated depth, allowing precise queue analysis	It shows <i>visible</i> <i>orders</i> , not private intent; it does not provide a guaranteed directional forecast, and the data volume, software needs, and real-time decision burden are substantial

8. Full microstructure research layer	MBO plus trade data, order-book event data, timestamp synchronization, economic-event labels, related-market data, and realistic execution records	Statistical modeling of fill probability, adverse selection, queue decay, order-book resiliency, market-impact estimates, regime detection, and execution optimization	The correct level for rigorous short-horizon research because it tests a defined rule against actual market events and costs	Expensive, technically difficult, vulnerable to overfitting, and structurally competitive with firms that have lower latency, co-location, and specialized execution systems
---------------------------------------	--	--	--	--

1. Tradingview

A tradingview chart should be your context layer, not your proof layer. It's excellent for identifying the larger structure of the market, showing the daily range, session high and lows, volatility expansion or contraction, and whether the market is rotating or trending. Tradingview provides futures quotes and charting access, but what you see depends on the symbol and data-subscription. Do not assume every chart is equivalent to a full exchange native feed.

The core limit is aggregation. A five-minute candle that closes green with 2,000 contracts of volume does not tell you whether buyers steadily lifted offers, sellers were absorbed, if a news event caused sudden repricing, a price sold off and reversed, the candlestick is simply a summary of all of those possible paths.

2. Intraday Resolution

The first step deeper is not necessarily DOM- it is simply breaking the chart into smaller-more individual observations. A daily bar can be decomposed into hourly bars, an hourly bar into five-minute bars, and a five-minute bar into one-minute, second, tick, range, or volume bars. This is useful because it lets you test *when* a condition occurred. For example, a strategy might find that a prior-day-high break behaves differently at the cash open than it does during lunch or after an economic release. The trade-off is that the finer bars can make random movement look meaningful, so the number of adjustable parameters must stay low.

3. Trades and Prints

Time-and-Sales data moves from bar summaries to the individual trades that formed those bars. From it, you can build volume profile, VWAP, volume-at-price distributions, trade-rate measures, and potentially classify activity as buyer or seller initiated. This lets you state facts such as "3,000 contracts traded at this price," "trading accelerated during this ten-second window" or "the market spent most of the session accepting prices in this range." It does not let you state without further evidence however, "an institution bought here and will defend the level." A print gives execution data, not participant identity or intention.

4. Best Bid and Ask

Top-of-Book data adds the best available buy and sell prices and their displayed sizes. It matters because a trader buys at the ask and sells at the bid; last price alone cannot accurately describe immediate execution conditions. From this layer, you can derive spread behavior, quote volatility, inside-market liquidity, and basic fill assumptions. But the best bid and ask are only the nearest displayed quotes. They do not show the entire order book, and the displayed size can disappear before you reach it.

5. Level 2 DOM

Market-by-Price depth adds a ladder of resting liquidity beyond the best bid and ask. CME's MBP feed aggregates quantity and number of orders at each price and limits the disseminated summary to the top 10 price levels; it cannot definitively identify an individual order or its queue position. This is valuable for execution awareness. You can examine where visible liquidity is concentrated, whether depth is un-usually thin, whether liquidity is being added or removed and whether your likely fill conditions are deteriorating. The error is treating the size on the DOM as a promise of support or resistance, remember: an order can be canceled, partially executed, refreshed, or used to influence other traders' expectations.

6. Footprints and Delta

Footprint charts are not a distinct exchange feed alone; they are a way of organizing executed-trade and quote data. They can display total volume at each price, bid volume, ask volume, delta, stacked imbalances, session delta, and related measures. Their greatest legitimate use is post-trade analysis and hypothesis formation. A footprint can show that aggressive market buying occurred at a high while price failed to continue higher. That observation may justify a testable hypothesis. It does not, by itself, prove "absorption," identify "smart money," or guarantee that price will reverse.

7. MBO Data

CME MBO is the deepest public view of the visible central limit order book. Rather than combining orders into one number at a price, it separates individual orders across the full depth of book and assigns anonymous identifiers and priority information that permit queue position analysis. MBO lets a trader or researcher study how visible liquidity behaves, whether an order is newly placed, partially filled, modified, canceled, replenished, or retaining priority. That is materially more informative than ordinary level 2 data.

I didn't elaborate on the eighth point because it is more of a compilation of each level specifically designed to give you the best view of each of the various tools. Ultimately each of these levels aren't here to show which one is better in terms of who is using it- so this isn't to say that someone trading candlesticks is worse than someone trading tape, simply put: Those going *deeper* will have access to a larger set of data and identifiers that allow those on a deeper level of data to see different market perspectives that a candlestick reader simply wouldn't see.

The basics of core market mechanics

Next we are going back to basics. Often I see people unable to identify exactly what each of these are and the following is the basis of how price moves and what causes it to move. This isn't complex mathematics by this point, we are simply going to be going over simple truths that pertain to the futures market itself.

To begin we ask the question: "What actually causes the price to move?" The answer is simple and direct:

"Futures price changes when aggressive orders consume the available resting liquidity at one price and transact at the next available price; resting limit orders supply liquidity and marketable orders demand immediacy. The order book is dynamic because liquidity can be added, canceled, modified, and executed."

Now, I will continue by defining common market factors:

- **Limit orders:** A limit order provides a price constraint but no certainty of execution; a market order prioritizes execution but NOT a specific final price.
- **Spread:** The bid-ask spread is the immediate cost of crossing the market, before commissions and fees.
- **Queue priority:** Being first at a price can matter because many CME products use price-time priority under FIFO, although CME supports several matching algorithms and the applicable method is contract-specific.
- **Displayed liquidity is conditional:** The DOM shows orders currently resting in the visible book, not durable commitments or participant intent.
- **Price Impact:** A trade's effect is not determined by size alone. The same 100-lot can be irrelevant in deep liquidity and meaningful in a thin, fast, or news-driven book.

Each and every one of these factors is just a small piece of building our "understanding" of a futures contract and how they work. We will proceed by diving into more core market mechanics, before studying indicators, order flow, quantitative models, or discretionary setups, you must understand the instrument being traded. A futures contract is not simply a bet on whether price will go up or down. It is a standardized agreement traded on an exchange with defined contract size, minimum price movement, trading hours, expiration, settlement procedures, margin requirements, and rules for how orders are matched.

The purpose of this section is to establish a factual baseline. These core mechanics do not tell us where price will go next, they simply explain what price represents, how trades occur, how profits and losses are calculated and which constraints any real trading strategy must survive. Beginning with Standardized Contracts:

Each futures product has a contract specification set by the exchange. The specification defines what the contract represents and how it trades. Before trading or analyzing the product, the trader should know the following:

- **Contract Multiplier:** The amount of dollar exposure represented by one point or index unit
- **Tick Size:** The smallest permitted price movement

- **Tick value:** The dollar profit or loss created by one tick for one contract
- **Trading hours:** When the contract is open and when liquidity is likely to be highest or lowest
- **Expiration month:** The month in which the contract stops trading or settles
- **Settlement method:** whether the contract is cash-settled or physically delivered
- **Margin requirements:** The collateral required to hold the position
- **Exchange and product rules:** The rules governing trading, settlement, position limits, and matching

For example, The Micro E-mini S&P 500 futures contract, MES, has a contract size of \$5 times the S&P 500 Index, a minimum price increment of 0.25 index points, and a tick value of \$1.25. The Micro E-mini Nasdaq-100, MNQ, has a \$2 multiplier, a 0.25-point tick, and a \$0.50 tick value. These specifications are not just minor details, as a trader cannot properly calculate risk without knowing each (well, they could, but it would be an unwise position to be in). Two contracts may look identical on a chart while exposing the trader to very different dollar amounts per point.

Moving on, we now address tick value and P&L. Futures profits and losses are calculated mechanically. If a position moves in the trader's favor, the account gains money; if it moves against the trader, the account loses money. The dollar value of that movement depends on the number of ticks moved, the tick value, and the number of contracts held, shown below:

$$\text{Profit or Loss} = \text{ticks moved} \times \text{tick value} \times \text{number of contracts}$$

For example, one MES contract moves \$1.25 per tick. A 10-tick move is therefore worth \$12.50 per contract before commissions, exchange fees, spread costs, and slippage. Ten MES contracts would make that same 10-tick move worth \$125 before costs. The risk of a trade can therefore be estimated before entry:

$$\text{Maximum planned loss} = \text{contracts} \times \text{stop distance in ticks} \times \text{tick value}$$

If a trader buys two MES contracts with a 12-tick stop, the planned price risk:

$$2 \times 12 \times \$1.25 = \$30$$

That is the starting point for position sizing. A setup is not "low risk" because it looks clean on a chart; it is low or high risk according to the defined loss amount relative to account equity. Next we will begin by addressing margin.

Margin is not risk

Futures are leveraged products. Margin is the amount of collateral a trader must deposit to open or maintain a position; it is not the maximum amount a trader can lose. The CFTC defines leverage as the ability to control a large dollar amount with comparatively little capital, while margin is the money or collateral deposited with a broker or clearing organization. This distinction is essential. A broker may allow a trader to control thousands or tens of thousands of dollars in notional exposure with a relatively small margin deposit. A small adverse price movement can therefore produce a large percentage loss

relative to the account balance. Futures margin can magnify returns, but it can also magnify losses and potentially lead to a margin call or as we call it: “blowing the account” or “being liquidated”.

A trader should never treat broker day-trading margin as a recommendation for size. It is merely the minimum collateral required by the broker under its rules. Prudent position size should be based on:

- **Account equity**
- **Defined maximum loss per trade**
- **Stop distance**
- **Tick value**
- **Expected volatility**
- **Liquidity condition**
- **Maximum daily and weekly drawdown limits**

The fact a trader *can* trade 10 contracts does not mean that 10 contracts is an appropriate amount of risk.

Unlike stocks, futures contracts expire. Equity-index futures, energy futures, metals, agricultural products, currencies, and interest-rate products all trade in specific contract months. A trader must know which contract is currently most liquid and when that liquidity will migrate to the next expiration. Retail charting platforms often display a continuous contract such as ES1! Or NQ1!, to make long-term chart analysis easier. A continuous chart is useful for visual context, but it is not one continuously traded contract. It is a constructed series that rolls from one expiring contract to the next.

This matters for research. Contract rollover can create artificial jumps, volume shifts, altered spreads, and changes in liquidity. A strategy tested on a continuous chart may produce misleading results if it does not account for:

- **The roll date**
- **The contract month actually traded**
- **Price adjustments applied by the chart provider**
- **Changes in volume and open interest**
- **Expiration-related behavior**
- **Different liquidity conditions in front-month versus deferred contracts**

A serious quantitative study or backtest of any sort should identify the exact contract and date range used, rather than assuming that continuous-chart data perfectly represents executable historical prices.

Settlement and mark-to-market

One defining feature of futures markets is daily mark-to-market. The exchange establishes an official settlement price for each contract, and open positions are credited and debited based on their gain or loss relative to the prior settlement price. CME notes that all futures contracts have daily mark-to-market prices and that the final settlement price is the same for all market participants in a given contract. At the end of the trading day, profits are credited and losses are debited from the trader's account. CME describes this process as marking positions to market, with accounts checked to ensure they maintain the required margin.

Mark-to-market serves several functions, allowing gains to realize daily rather than allowing them to accumulate indefinitely. It assists firms in managing counterparty risk, determines whether an account has sufficient margin to maintain the open position, and triggers additional margin requirements if losses reduce the account equity below required levels.

The settlement price is not always identical to the final last-traded price visible on a chart. Exchanges use published methodologies based on trading activity during designated settlement periods or other defined procedures. CME states that settlement methodologies are disclosed in contract specifications and the exchange rulebook. For equity-index futures for example, CME may determine settlement using a specified window of Globex trading activity. The exact procedure can vary by contract and should be checked in the applicable product rules.

Cash settlement and delivery

Futures contracts generally settle in one of two ways:

- **Cash settlement:** The contract settles financially based on a reference value or final index value. Equity-index futures such as ES, NQ, MES, and MNQ are commonly cash-settled.
- **Physical delivery:** The contract may require delivery of the underlying commodity if held through the relevant delivery period. This is more relevant to many agricultural, energy, and metal futures products.

A retail trader who day trades futures may rarely hold a position near expiration, but this does not make delivery rules irrelevant. Contract expiration can create changes in liquidity, volatility, spread behavior, and position management. Traders must know the final trading day and applicable delivery or settlement rules before carrying a position forward.

Trading sessions and liquidity

Futures markets often trade for most of the week through electronic platforms such as CME Globex. However, "open" does not mean liquidity is constant. CME Globex hours vary by product, and product specifications should always be checked rather than assumed. Gold futures, for example, list CME Globex trading from Sunday through Friday with a daily maintenance break. For many U.S. index-futures, the most active period often overlaps with regular U.S. cash-equity hours. CME identifies

regular trading hours for U.S. equity-index futures as 8:30 a.m. to 4:00 p.m. Central Time, though market schedules can change for holidays and other special circumstances.

Time matters because liquidity, volatility, spread, and order-book depth vary substantially across the session. A strategy that appears stable during the cash open may behave completely differently overnight or during the midday period.

Important session distinctions include:

- **Overnight trading versus U.S. regular trading hours**
- **The U.S. cash-equity open**
- **Scheduled macroeconomic releases**
- **Treasury auctions and central-bank announcements**
- **The midday liquidity decline**
- **The cash-equity close**
- **Futures settlement windows**
- **Contract rollover periods**
- **Holiday and shortened-session trading**

A strategy must therefore be evaluated not only by price pattern, but also by when it trades. Time of day is market variable, not background information.

The Order Book

Futures markets operate through a central limit order book (often referred to as the COB, or LOB). Participants submit orders to buy or sell at specified prices, and the exchange matches compatible orders according to the products matching rules.

At any moment, the order book contains visible resting buy and sell orders:

- **Bid:** The highest currently displayed price at which someone is willing to buy
- **Ask or offer:** The lowest currently displayed price at which someone is willing to sell
- **Bid-ask spread:** The difference between the best bid and the best ask
- **Depth:** Resting displayed quantity at prices beyond the best bid and ask
- **Liquidity:** The ability to trade a given quantity with limited price movement and reasonable execution quality.

A limit buy order resting below the market provides potential liquidity to sellers. A limit sell order resting above the market provides potential liquidity to buyers. A marketable buy order consumes available sell liquidity at the ask; a marketable sell order consumes available buy liquidity at the bid. Price moves when available liquidity at a price is consumed or withdrawn and trades occur at the next available price. This is a more accurate description than saying that price moves simply because “buyers outnumber sellers”. Every completed trade has both a buyer and a seller. What changes price is the urgency of the orders, the location of resting liquidity, and the willingness of participants to transact at new prices.

Limit orders and Market orders

A trader generally chooses between providing liquidity and demanding liquidity. A limit order specifies the worst acceptable price. A buy limit order may execute at the specified price or lower; a sell limit order may execute at the specified price or higher. The advantage is price control. The disadvantage is uncertainty: the order may not fill, may fill only partially, or may fill just before price moves against it.

A market order prioritizes execution rather than price. A market buy order executes against the best available asks; a market sell order executes against the best available bids. The advantage is speed and a higher likelihood of immediate execution. The disadvantage is that the final fill may be worse than expected when liquidity is thin or price is moving quickly. Ultimately this creates a basic trade off, with limit orders reducing the cost of crossing the spread but introducing non-fill and adverse selection risk, and market orders creating an increase in certainty of execution but pay the spread and may incur slippage.

A quantitative model that assumes perfect fills at every limit price or stop price is not modeling a realistic trading environment. This goes for any backtest or test of any type on a strategy that is expected to make profit.

Spread and execution cost

The bid-ask spread is the first execution cost that a trader encounters. If the best bid is 5,000.00 and the best ask is 5,000.25, a trader who buys immediately pays 5,000.25 while a trader who sells immediately receives 5,000.00. The position begins with a one-tick disadvantage before commissions, exchange fees, and slippage.

The all-in cost of a trade can be described as:

$$\text{All cost} = \text{commissions} + \text{exchange fees} + \text{spread cost} + \text{slippage} + \text{market impact}$$

Commissions and fees are explicit costs charged by the broker, exchange, clearing firm, and regulators. Spread cost results from crossing from bid to ask or ask to bid. Slippage is the difference between the expected price and the actual execution price, and market impact is the price movement created or worsened by the trader’s own order. For a small retail trader in a highly liquid contract, market impact may be negligible during normal conditions. During major news events, rollover, thin overnight sessions, or sudden volatility, the same order size can experience materially worse execution.

A strategy should be judged on net results after all costs. A backtest that produces a small gross profit per trade may be untradeable once real execution is included.

Queue Order and Priority Fills

When multiple participants place orders at the same price, the exchange must determine which order receives the next fill. Many CME products use price-time priority, commonly described as first-in, first-out. Under FIFO, better-priced orders receive priority, and among orders at the same price, earlier orders generally receive priority. CME describes FIFO matching as filling outright orders at the same price level according to time priority. CME also supports multiple matching algorithms, including FIFO and allocation-based methods, depending on the product and order type. The exact matching rules must be verified for the specific market being traded.

Queue position matters most for traders who rely on resting limit orders. Seeing 500 contracts displayed at a price does not mean a 10-contract limit order placed behind them will be filled when 10 contracts trade there. The trader may be hundreds of contracts back in the queue, other orders may receive priority, liquidity may be canceled, or the market may trade through the price too quickly for a fill. This is one reason MBO data is valuable for research: it can help reconstruct visible individual orders and queue behavior. It does not, however, eliminate fill uncertainty or guarantee that a limit-order strategy is viable.

Remember, displayed liquidity is conditional, the DOM and MBO reveal visible resting liquidity, but the displayed orders should not be mistaken for permanent commitments. And order can be:

- **Added**
- **Modified**
- **Partially filled**
- **Fully filled**
- **Canceled**
- **Replaced at another price**
- **Replenished after execution**

The presence of a large bid does not prove support, and a large ask does not prove resistance. It only proves that visible liquidity exists at that moment. A resting order may represent genuine interest, a hedging instruction, a market-making process, an execution algorithm, or an order that will be canceled before it trades. This does not make order-book data useless. It means it must be interpreted correctly. The trader can observe changes in visible liquidity, not private intent. The appropriate use of the data is to measure execution conditions, formulate hypotheses, and test whether certain order-book states have predictive or execution value after costs.

What these core mechanics establish

These mechanics establish several facts:

- **Futures are standardized leveraged contracts with fixed tick values and defined expiration and settlement rules.**
- **Margin is collateral, not a measure of maximum risk.**

- Price changes through the interaction of aggressive orders and available resting liquidity.
- A chart records a summary of market activity, while deeper data provides more detailed records of traders and orders.
- Spread, commissions, slippage, fill uncertainty, and market impact are part of every real strategy.
- The same price pattern can behave differently across sessions, volatility regimes, contract months, and liquidity conditions.
- Visible order-book data can improve observation, but it cannot reveal participant identity, private intent, or future certainty.

Price, Volume, and Order Flow

Price, volume, order flow, and open interest are related, but they are not interchangeable. Each records a different part of market activity, and none independently reveals future direction, participant identity, or intent. Below is a representation of each, what they record, what they can help measure, and what each cannot prove alone:

Measure	What it records	What it can help measure	What it cannot prove alone
Price	The price at which trades occurred	Trend, range, returns, volatility, and location within a session or range	Why the trade occurred or where price will go next
Volume	Number of contracts traded	Participation, trade intensity, volume-at-price, VWAP, and activity by time of day	Whether activity is inherently bullish or bearish
Bid/ask volume	Executed volume classified at or near bid versus ask	Aggressive buying/selling estimates, delta, and short-term imbalance	“Smart money” direction, participant identity, or a guaranteed reversal
Open interest	Futures contracts still open at the end of a trading day	Broader participation, contract-month activity, and longer-term changes in outstanding exposure	Intraday direction, motive, or which side of the trade is correct

A completed futures transaction always has both a buyer and a seller. For that reason, saying “there were more buyers than sellers” is inaccurate when referring to completed volume. A more precise statement is that buyers were more aggressive: they used marketable buy orders to lift offers, causing transactions to occur at progressively higher prices. Sellers are more aggressive when they use marketable sell orders to hit bids and transact at lower prices. This distinction is important because order flow describes how trading occurred, not necessarily what will happen next. Heavy buying at the ask may lead to continuation, exhaustion, reversal, or no meaningful follow-through. The outcome depends on the

surrounding liquidity, the response of resting participants, the broader market regime, and the execution environment.

Open interest

Open interest is the number of futures contracts that remain open at the end of a trading day. It differs from volume: volume counts every contract traded during a session, while open interest counts contracts that remain outstanding after positions are opened, closed, or transferred. A market can trade high volume while open interest changes very little. That may occur when traders are transferring existing positions rather than creating new exposure. Conversely, open interest increases when new positions are opened and declines when existing positions are closed. Because each futures contract always has a long and short side, rising open interest does not prove that bullish participants are “taking control,” nor does falling open interest prove bearish conviction.

Open interest is most useful as delayed context. It can help study participation across contract months, liquidity migration during rollover, and larger changes in outstanding exposure. It is not a real-time intraday execution signal and should not be treated as one.

What Data Cannot Reveal

Market data records visible events. It does not provide direct access to motive, identity, or future behavior. This distinction matters because much retail trading language turns observable activity into a confident story about “institutions,” “smart money,” or trapped participants.

Market data cannot, by itself:

- **Identify the intent of an anonymous order or execution**
- **Prove that a large trade came from one participant rather than multiple matched participants**
- **Prove that a large print represents directional conviction rather than hedging, a spread, liquidation, or an execution algorithm**
- **Reveal whether a visible limit order will remain, cancel, modify, or execute**
- **Show private information, unexpressed future orders, or activity outside the visible central order book**
- **Establish that a pattern seen on one chart will persist in future conditions**
- **Establish that one side of the trade is “smart money” and the other side is “dumb money”**

This does not make the data useless. It defines the correct use of the data. “Aggressive buying increased at the session high, but price failed to continue higher” is an observation. “Institutions trapped retail buyers and will now force the market lower” is an unverified explanation. The first statement can become a hypothesis and be tested. The second usually cannot be tested because it assumes information that the data does not contain. Quantitative analysis is valuable precisely because it forces the trader to separate what was observed from what was inferred.

Time, Volatility, and Regime

Markets do not behave identically across all periods. A pattern that appears profitable in one sample can fail when liquidity, volatility, participants, macroeconomic conditions, or market structure changes. This is known as regime dependence or nonstationarity. A strategy should not only be asked, “Did it work historically?” It should also be asked, “Under which conditions did it work, and under which conditions did it fail?”.

Important regime variables include:

- **Low-volatility versus high-volatility sessions**
- **Trending versus rotational or mean-reverting conditions**
- **Regular U.S. trading hours versus overnight conditions**
- **Deep, stable order books versus thin or rapidly changing books**
- **Normal sessions versus CPI, payrolls, FOMC, Treasury-auction, or major earnings days**
- **Contract rollover and expiration periods**
- **Risk-on versus risk-off macroeconomic conditions**
- **Stable versus unstable correlation between equity-index futures, bonds, currencies, and volatility markets**

The cash-equity open, an FOMC statement, the lunch period, and the last hour of regular trading should not be treated as statistically identical. Liquidity, spread, volatility, participation, and the chance of adverse execution differ across these periods. If a strategy works only in ES, only during regular trading hours, and only when volatility is above a defined level, that does not necessarily invalidate the strategy. It identifies its potential operating conditions. The mistake is claiming that a conditional result is a universal law of markets.

Expected Value, Not Win Rate

The goal of a strategy is not a high percentage of winning trades. The goal is positive expected value after commissions, exchange fees, spread costs, slippage, and other realistic execution assumptions. (shown below)

$$Expectancy = (P_w \times \bar{w}) - (P_l \times \bar{L}) - C$$

Where:

- P_w is the probability of a winning trade
- $\bar{w}$ is the average winning trade
- P_l is the probability of a losing trade
- $\bar{L}$ is the average losing trade

- C is the average all-in cost per trade

Since $P_l = 1 - P_w$, win rate is only one component of a system's result. A strategy can win 40 percent of the time and still be profitable if average winners are sufficiently larger than average losses. A strategy can win 90 percent of the time and still lose money if its rare losses are large enough. For example, consider a strategy that wins 40 percent of its trades, makes \$300 on average when it wins, loses \$100 on average when it loses, and incurs \$10 in total average trading cost:

$$(0.40 \times 300) - (0.60 \times 100) - 10 = 50$$

Its estimated expectancy is \$50 per trade. That does not mean the next trade should make \$50. It means the strategy would be expected to average roughly \$50 per trade over a sufficiently large sample if the measured conditions remain stable. Expectancy is an estimate rather than a guarantee. It can disappear when costs rise, volatility changes, liquidity worsens, the market regime shifts, or a historical result turns out to have been random chance.

Drawdown and Survival

Positive expectancy does not guarantee that a trader survives long enough to benefit from it. A positive strategy can still fail if position sizing is too large, execution differs from testing assumptions, or a normal losing streak exceeds the trader's financial or psychological capacity. A drawdown is the fall from an account's previous peak value to a later low. Drawdowns are normal in any uncertain process. The relevant question is not whether a drawdown occurs, but whether it is within the expected historical range and whether the account can withstand it without forcing a change in behavior.

Risk of ruin is the risk that losses reduce an account to a level where the trader cannot continue trading or cannot trade at a meaningful size. This risk rises when a trader:

- **Risks too much account equity on one trade**
- **Increases size after a winning streak**
- **Attempts to immediately recover losses through revenge trading**
- **Treats several correlated futures positions as diversification**
- **Uses stop distances that do not fit current volatility**
- **Trades high-impact scheduled events without allowing for worse fills**
- **Bases position size on broker margin rather than defined maximum loss**

Risk management does not create edge. It preserves the ability to exploit an edge if one exists. A trader risking 20 percent of account equity per trade is not displaying greater conviction; they are allowing position sizing to dominate the outcome of the strategy. The next step is not to search for a flawless setup. It is to take a market observation and state it precisely enough that it can be tested, rejected, or supported.

For example a weak claim might be: *"Price rejected the high after buyers got trapped so it should reverse."*

A testable hypothesis is different:

“During ES regular trading hours, if price tests the prior-day high, cumulative delta makes a 30-minute high, and price fails to close above the prior-day high, enter short only after a five-minute close below the local swing low. Use a fixed stop and target, apply realistic commissions and slippage, and test the rule separately during low- and high-volatility sessions.”

The testable version may fail, which is exactly why it is useful. A hypothesis that cannot fail cannot be meaningfully evaluated.

Every quantitative trading hypothesis should specify:

- **The exact contract and contract-roll method**
- **The dates studied**
- **The trading session and time zone**
- **The data source and data granularity**
- **Exact entry, exit, stop, target, and time-based exit rules**
- **Position sizing and maximum risk assumptions**
- **Commission, fee, spread, slippage, and partial-fill assumptions**
- **In-sample and out-of-sample test periods**
- **Number of adjustable parameters**
- **Performance measures, including expectancy, drawdown, win rate, profit factor, and sensitivity to execution costs**

A trading claim remains unproven until it survives defined rules, realistic costs, out-of-sample testing, and multiple market regimes. That is the proper transition from observing futures markets to conducting quantitative analysis: not finding certainty, but systematically rejecting claims that do not survive measurement.

From Observation to Measurement

At this point, we understand that market data records events rather than intentions. Price tells us where the trades occurred. Volume tells us how much traded. Order-flow and order-book data provide more detail about executions and visible liquidity. None of these facts automatically tell us what will happen next. The purpose of quantitative work is to take an observation and turn it into a claim that can be tested. The aim is not to create a story that explains a chart after the fact. The aim is to state a rule before viewing the result and then determine whether the rule has any value beyond chance.

Ultimately the goal is to form a useful hypothesis. A useful hypothesis has five properties:

- **Specific:** Another person can identify the same entry and exit without guessing.
- **Falsifiable:** A test can produce a result that rejects the claim.
- **Measurable:** The required data and calculations are available.

- Economically realistic: It includes costs, fills, and the actual instrument traded.
- Limited: It does not depend on dozens of discretionary exceptions or adjustable settings.

The job of a trader is not to defend a favorite concept. It is to test whether the concept improves decisions under defined conditions.

Defining a Trade Before Testing

A Backtest cannot evaluate a vague setup. Before testing begins, the trader must write the full trade definition. This should happen before looking at results, not after a promising pattern is found.

Every trade definition should state:

- Instrument: The exact contract being tested, such as ES, MES, NQ, MNQ, CL, or GC.
- Data period: The beginning and ending dates, including how contract rollover is handled.
- Session: The hours during which entries are permitted, stated in a clear time zone.
- Entry trigger: The precise condition that opens a position.
- Direction: Whether the rule is long-only, short-only, or both.
- Entry method: Market order, limit order, stop order, or another defined execution method.
- Stop-loss rule: The exact price, tick distance, volatility measure, or structural condition that exits a losing trade.
- Profit-taking rule: A fixed target, trailing rule, time exit, opposing signal, or other defined method.
- Time exit: Whether open positions are closed at a fixed time or carried overnight.
- Position size: The number of contracts or a defined risk-based sizing formula.
- No-trade conditions: Scheduled events, low-liquidity periods, holidays, rollover days, or maximum-loss limits.
- Cost assumptions: Commissions, exchange fees, spread, slippage, and partial-fill assumptions.

A rule such as “enter at a fair-value gap after a market-structure shift” is not sufficiently defined. It requires the trader to define what counts as a fair-value gap, which swing establishes structure, what timeframe is used, what invalidates the setup, and when the trade is no longer valid. If two people cannot code the same rule or manually label the same events with high agreement, the rule is too subjective to evaluate honestly. Discretion may still have value, but it should be identified as discretion. It should not be advertised as a fully repeatable mechanical strategy if the real decision is being made by the trader’s judgment.

In-Sample and Out-of-Sample Testing

A strategy can appear excellent on the same data used to create it. This does not prove that it has an edge. It may simply mean that the trader found a rule that fits historical noise. To reduce this risk, historical data should be divided before research:

- In-sample data: The period used to develop the hypothesis and select limited parameters
- Out-of-sample data: A separate period that is not used during development and is reserved for evaluation.
- Forward testing: New data collected after the rule is finalized ideally through live simulation or small real size

For example, a trader might develop a rule on ES data from 2021 through 2023, reserve 2024 for out-of-sample testing, and then forward-test it during 2025. The exact dates are less important than the discipline: the out-of-sample period must not influence the design of the strategy. If a rule loses money out of sample, changing its parameters until it works again does not restore the original validation. It turns the supposedly independent sample into new in-sample data. The revised rule must then be tested on a fresh, untouched period. Forward testing is valuable because it exposes the strategy to present-day conditions, actual data handling, realistic decision-making, and live execution. It will not prove permanent profitability, but it is stronger evidence than repeatedly optimizing historical charts.

Overfitting and Data Mining

Overfitting occurs when a strategy is designed so closely around historical data that it captures random variation instead of a durable relationship. It is one of the largest risks in retail quantitative work because modern platforms make it easy to test thousands of indicators, timeframes, filters, stop sizes, targets, and entry variations. A strategy is more vulnerable to overfitting when it contains:

- Numerous indicators or filters
- Many adjustable parameters
- Exact values selected only because they produced the best past result
- Separate exceptions for losing examples
- Very small sample sizes
- Performance concentrated in a short period
- Results that disappear after modest changes to stop size, target, time window, or costs
- A complicated explanation with no plausible market mechanism

The more variations a researcher tests, the more likely one will look profitable by chance. Finding one impressive result among hundreds of attempts is not the same as discovering an edge. A more credible strategy is usually simple enough to explain in one paragraph, stable under small changes, profitable across multiple independent samples, and not dependent on a single exact parameter. If a strategy works only with a 17-minute opening range, a 1.73-standard-deviation filter, a 13-tick stop, and a 29-tick target – but fails immediately when any input changes slightly – it is likely fragile.

The appropriate response to an attractive backtest is skepticism. Ask what assumptions created the result, whether the rule would have been known in advance, whether the test includes realistic execution, and whether the effect survives outside the original sample.

Baselines and Comparisons

A result is not meaningful merely because it made money. It must be compared with an appropriate baseline. For a directional intraday rule, possible baselines include:

- Random entry at the same time of day with the same stop and target
- A rule that enters in the opposite direction
- A simple opening-range or session-based entry rule
- Buy-and-hold or short-and-hold where appropriate
- A rule with the same holding period but no indicator or order-flow condition
- The same rule applied without the proposed filter

The purpose is not to prove that random entry is useful. It is to determine whether the feature being claimed — delta divergence, a prior high, an order-book imbalance, VWAP location, or another condition — actually improves the outcome beyond a simpler alternative. For example, suppose a trader claims that a stacked footprint imbalance predicts continuation. The correct test is not simply to display a profitable equity curve after finding historical examples. The trader should compare the results of trades with the imbalance against otherwise similar trades without it. If the two outcomes are indistinguishable after costs, the footprint condition is not adding demonstrated value.

Measuring a Strategy

A profitable-looking equity curve is not enough. A strategy should be described using a group of measurements because each reveals a different weakness.

Measure	What it tells us	What it can hide
Net profit	Total money made or lost in the sample	Position size, volatility, and risk taken to produce it
Expectancy	Average expected result per trade after costs	Whether the estimate is stable or based on too few trades
Win rate	Percentage of trades that won	Average win, average loss, and tail risk
Average win / loss	Typical payoff asymmetry	How outcomes are distributed around the average

Profit factor	Gross profit divided by gross loss	Whether one unusually large trade created the result
Maximum drawdown	Largest peak-to-trough account decline	Whether the sample was long enough to reveal worse drawdowns
Trade count	Amount of evidence available	Whether trades are independent or clustered
Exposure time	How long capital is at risk	Overnight, event, and gap risk
Cost sensitivity	How results change with worse execution	Whether a small cost increase destroys the edge

The key question is not, “How much did the strategy make?” It is, “What return did it produce for the risk, drawdown, time, and execution assumptions required?” A strategy that earns \$10,000 while experiencing a \$9,000 drawdown may be less useful than one that earns \$6,000 with a \$1,000 drawdown. Likewise, a strategy that makes a small profit before costs but loses after one additional tick of slippage is not robust enough for live use.

Robustness Testing

After a strategy produces a promising result, the next objective is not to optimize it further. It is to attempt to break it. A robust strategy should be examined under reasonable variations:

- Increase assumed slippage by one or more ticks.
- Increase commissions and fees to a conservative estimate.
- Change the entry window modestly.
- Change the stop and target modestly.
- Test long and short sides separately.
- Test separate years and market regimes.
- Test regular trading hours and overnight conditions separately.
- Remove the best-performing month or day.
- Exclude high-impact news days.
- Compare the main contract with its micro equivalent where execution assumptions permit.
- Randomize trade order to study the distribution of drawdowns.
- Check whether a small number of outlier trades account for most profits.

A strategy that remains reasonable after these tests is not proven forever. It is simply more credible than one that collapses when a parameter changes from 20 ticks to 21 ticks. Robustness should be valued over the highest historical return. A lower-return strategy that remains positive under realistic variations is generally more useful than a highly optimized strategy that works only under one narrow set of assumptions.

A Research Journal

Every quantitative project should have a written research journal. Without one, a trader is likely to forget how many ideas were tested, how often parameters were changed, and whether a current result was actually discovered before or after examining the data. For each hypothesis record:

- The original observation
- The precise rule written before testing
- The source and granularity of data
- The period tested and rollover treatment
- All parameters and why they were selected
- Assumed costs and fill model
- In-sample and out-of-sample results
- Robustness-test results
- Charts and trade examples, including losing examples
- The decision: reject, revise, forward-test, or retire

A journal created accountability. It prevents the trader from remembering only successful ideas and ignoring the hundred variations that failed. It also makes it possible to distinguish genuine improvement in research from repeated redesign after seeing disappointing results.

Transition to Application

The next stage is to apply this framework to futures-specific research. The objective is not to identify a universal setup, nor to claim that one data type is superior to every other. It is to choose one observable condition, define it exactly, determine the appropriate data resolution, and test whether it improves a trading decision after costs and across conditions.

A practical research sequence is:

- Select one liquid contract and one clearly defined session.
- Identify one observable condition, not a full collection of indicators.
- Create exact entry, exit, and risk rules.

- Test the rule with conservative cost assumptions.
- Compare it against a simple baseline.
- Separate development data from evaluation data.
- Stress-test the rule across market regimes and execution assumptions.
- Forward-test only after the rule is fixed.
- Scale size only after live results support the original research.

This is the point where trading becomes less about finding an explanation for a chart and more about building a repeatable decision process.

That repeatable decision process still needs a foundation underneath it before it can be built. Every section before this one has quietly used statistical reasoning without naming it. "In-Sample and Out-of-Sample Testing" assumes you already know what's allowed to change across time and what isn't. "Overfitting and Data Mining" is, underneath the plain-English framing, a warning about statistical generalization. And the \$50-per-trade result at the center of "Expected Value, Not Win Rate" is a sample mean — and every sample mean carries uncertainty whether you calculate it or not. Before any order-flow metric enters this paper, four ideas need to be made explicit, because you can't evaluate any quantitative metrics with any honesty if you can't first tell the difference between a real relationship and a coincidence that happened to survive your testing window.

A process is stationary if its basic behavior (its mean, variance, and how it relates to its own past) doesn't depend on when you sample it. A coin flip is stationary, the odds of heads are the same on Monday as they are during a recession. Raw futures price is not. Plot NQ over any trending week and the average price on Monday is not the average price on Friday, the level of price itself has drifted. That's not a minor technicality. The tools in the next three sections all assume you are sampling repeatedly from something that isn't moving the goalposts underneath you. Apply them directly to a price level and you get nonsense, real correlations and price relationships that vanish the following week, and false ones that look rock-solid simply because two trending series correlate with each other even though they have nothing to do with one another. This is a well-known trap called spurious regression, and it's the most common way retail "quant" analysis fools itself without a single instance of curve-fitting.

The practical fix is to stop feeding levels into anything and start feeding differences or bounded ratios instead. Returns (the change from one price to the next) are far closer to stationary than the price series they are computed from. For instance the average return over a given week sits close to zero and doesn't drift the way price does, even during a strong trend. That's exactly why order-flow features orderflow imbalance, order book imbalance, microprice deviation from mid price, each are more tractable research than the price itself. They are built to oscillate around a fixed reference point (usually zero) rather than to wander. That doesn't make them perfectly stationary, their variance changes across sessions. For instance:

A quiet Tuesday afternoon and the ten minutes after a CPI print are not drawn from the same distribution, which is its own form of nonstationarity (heteroskedasticity) that later robustness testing has to account for. But "sometimes noisier, sometimes calmer" is a survivable problem for a model. "The average value itself keeps drifting to a new number" is not. Before trusting any relationship between a feature and forward returns, the first honest step is a stationarity test on the feature series itself, such as the Augmented Dickey-Fuller test. If it fails, nothing built on top of that feature is trustworthy no matter how good the backtest looks.

Autocorrelation: Does the Past Actually Predict the Future?

Once you're working with something closer to stationary, the next question is whether the past of that series tells you anything about its future at all. The relationship has a name, Autocorrelation, the correlation between a series and a lagged copy of itself, meaning the same series compared against itself some number of steps earlier. At some lag k :

$$p(k) = \text{Cov}(X_t, X_{t,k}) / \text{Var}(X_t)$$

Where:

- X_t is your feature's value right now (this minute's order flow imbalance for instance)
- $X_{t,k}$ is the same feature's value k steps later, one minute later if $k=1$, five minutes later if $k=5$
- $\text{Cov}(X_t, X_{t,k})$ asks one plain question: when the value right now sits above its own average, does the value k steps later also tend to sit above its average or below it? Both high together (or both low together) pushes this positive. One high lining up with the other low pushes it negative. No consistent pattern averages out near zero.
- $\text{Var}(X_t)$ is that same calculation done against itself at zero lag, in more plain terms, how far the series typically sits from its own average. Dividing by it does one job, it rescales a raw covariance number, stuck in whatever messy units your feature happens to use, into a clear number between -1 and 1 that you can compare across completely different features on the same scale.

So in practice, picture tracking one-minute order flow imbalance. If a strong positive reading tends to get followed a minute later by another strong positive reading (and a negative reading by another negative one), that's positive autocorrelation at lag 1. The feature has short-term momentum, and this minute's reading is telling you something real about the next one. If a strong positive reading tends to get followed by a swing back toward negative that's negative autocorrelation (the feature tends to snap back on itself). And if this minute's reading has no consistent relationship to the next one at all, autocorrelation sits near zero, and any "pattern" you think you're seeing is decoration sitting on top of noise.

There's a trap here that only shows up if you never test it formally. Suppose price does nothing but bounce between the bid and the ask, with no informed trading, no real signal, just routine order flow crossing the spread in both directions. Every uptick (a trade at the ask) is mechanically likely to be followed by a downtick (a trade at the bid), and vice versa, purely because of where the trade executed relative to the quote. Not because anyone's view of the value changed though. This produces a strong negative autocorrelation in trade-to-trade price changes that has nothing to do with a tradable mean-reversion edge; it's microstructure noise wearing the costume of a signal. This mechanical bounce has a formal name and a full model, and we will continue to discuss that later.

For now, the operating rule is: before attaching a trading rule to "the market tends to reverse after X ," run the autocorrelation function on your actual feature at the actual lag you intend to trade, and check whether it survives once genuine signal is separated from this kind of structural bounce. A formal test like the Ljung-Box test is the standard one to tell you whether autocorrelations across a set of lags are jointly distinguishable from what pure noise would produce. Skipping that step and going straight to "it looks like it reverses on the chart" is the same unfalsifiable pattern-reading this paper's opening spent several pages debunking. The only difference is it now has an indicator attached instead of a fibonacci grid.

Assume, for a moment, you've done the work: you've found a feature that's reasonably stationary, and it shows a genuine, non-spurious relationship with forward returns at the lag you care about. You build a rule around it and backtest it. It wins 40% of the time, averages \$300 on a winner, loses \$100 on a loser, and costs \$10 all-in per trade (the exact numbers already used in "Expected Value, Not Win Rate"). That gave you \$50 per trade. The question this section exists to answer is blunt: how much do you trust that \$50?

Every individual trade outcome in that sample bounces around its own average, some winners, some losers, with a standard deviation you can calculate directly from the win/loss numbers above: roughly \$196 per trade. That's the noise in any single result. What you actually care about isn't one trade, it's the average across your whole sample, and the average of many noisy draws is itself less noisy than any individual draw, but only in proportion to the square root of how many draws you have, not in direct proportion to the count. That shrinkage rate has a name: the standard error of the mean, calculated as the standard deviation of individual outcomes divided by the square root of the sample size ($SE = \sigma / \sqrt{n}$). This is the Central Limit Theorem doing its actual, practical job. No matter how skewed the underlying win/loss distribution looks, the average of enough independent draws from it starts behaving like a normal distribution, and that's what lets you say anything precise about how far your measured \$50 might sit from the true, long-run number.

Run that arithmetic at three sample sizes and the standard error drops from roughly \$35.80 at 30 trades, to \$11.30 at 300 trades, to \$3.60 at 3,000 trades. Thirty trades is not a small caveat here, it's the difference between a strategy and a coin flip that happened to land well. This is the concrete, arithmetic version of something "Overfitting and Data Mining" already told you to be suspicious of: a "very small sample size" isn't just aesthetically less convincing, it means the standard error is wide enough that your headline number could plausibly be almost anything within a very wide range. The next section shows exactly how wide.

Hypothesis Testing: p-values, Confidence Intervals, and the Cost of Testing Everything

Standard error becomes useful once it's converted into an actual yes-or-no test. The question "is this \$50 real" is formally the question: if the true edge were actually zero, how surprising would it be to see a sample average this large from noise alone? That's a hypothesis test. The null hypothesis is that the strategy has no true edge, making the real average over infinite trades, is zero. The test statistic, a t-statistic, is just the measured average divided by its standard error:

$$t = (\bar{x} - 0) / SE$$

Telling you how many standard errors your result sits away from zero.

The p-value converts that t-statistic into a probability: specifically, the probability of seeing a result at least this extreme if the null hypothesis were actually true. A common misreading is worth killing early- A p-value is not the probability the strategy works, and it is not the probability the null hypothesis is true. It only answers "how surprising would this look if there were nothing here at all."

A confidence interval expresses the same evidence as a range instead of a single verdict:

$$\bar{x} \pm t^* \times SE$$

Where t^* is the critical value from the t-distribution at the desired confidence level, roughly 1.96 to 2.05 across the sample sizes below. The table applies both formulas together:

Sample Size (n)	Standard error	T - statistic	P - value	95% confidence interval	Verdict
30 trades	\$35.80	1.40	≈ 0.17	-\$23 to \$123	Not distinguishable from zero
300 trades	\$11.30	4.42	< 0.0001	\$28 to \$72	Real, check economic size next
3,000 trades	\$3.60	13.97	< 0.0001	\$43 to \$57	Statistically solid

Same headline \$50, three completely different levels of evidence. At 30 trades, a strategy with genuinely zero edge would still throw up a result this good roughly one time in six from sampling luck alone, not a bar any serious claim should be allowed to clear. The confidence interval tells the same story in a more useful shape, because it also communicates how big the edge plausibly is, not just whether it's nonzero: at 30 trades the interval comfortably includes zero and even includes losing money; at 3,000 trades it's narrow enough to actually plan around.

Two honest caveats belong here, not buried in a footnote. First, every calculation above assumes the trades are independent, identically distributed draws (the exact stationarity assumption from the start of this chapter). If your trades cluster, several losers in a row because they all fired during the same adverse regime, or the win rate genuinely shifts month to month, the real standard error is wider than the formula admits, and a result that looks significant at 300 trades under this naive assumption might not be. This is precisely what "Measuring a Strategy" already flagged under trade count: whether trades are independent or clustered changes what the count is actually worth. Second, statistical significance is not economic significance. A p-value under 0.0001 tells you an edge almost certainly isn't zero. It says nothing about whether that edge survives the spread, commissions, and slippage the all-in cost formula already requires. A statistically bulletproof \$2-per-trade edge is still not a strategy you can trade.

There's one more cost of testing that "Overfitting and Data Mining" already warned about from a different angle, without a number attached. Test one rule and clear a 5% significance threshold, that's meaningful. Test 100 unrelated rules (100 combinations of indicators, timeframes, and filters) at that same 5% threshold, and simple arithmetic says you should expect roughly five of them to look "significant" from pure chance alone, with better than a 99% chance that at least one clears the bar by luck across all 100 attempts. That's the quantified version of a sentence already in this paper: the more variations a researcher tests, the more likely one will look profitable by chance. The practical fix, when testing many variants, is to hold yourself to a far stricter threshold than 5% per test, roughly in proportion to how many variants you actually tried, and to treat any single winning configuration found this way as a hypothesis for the out-of-sample and forward-testing stages already described, never as a conclusion on its own.

These ideas answer one question honestly: is what you're looking at a real, testable relationship, or noise that happened to survive your particular sample? That has to be settled before a single order-flow metric enters this paper, because every metric from here forward: microprice, OBI, OFI, VPIN, are only as trustworthy as the statistical discipline applied to it. It's also worth naming what none of this explains: why any of these relationships would exist in the first place, why order flow should predict short-horizon price movement at all, or why a market maker would ever let a detectable imbalance

persist long enough to trade against. That's a theory question, not a statistics question, and it's where this paper goes next.

Every metric still ahead in this paper, microprice, order book imbalance, order flow imbalance, VPIN, is a way of measuring who is trading and why. None of that is worth building until you understand why a bid and an ask exist as two different numbers in the first place, and why the gap between them changes size on its own, without anyone touching a chart pattern. There are three explanations for that gap, developed by three different people in the 1980s, and they stack on top of each other. Each one also quietly answers a question this paper already raised without naming a mechanism: the bounce mentioned in "Autocorrelation: Does the Past Actually Predict the Future?", the "same 100-lot, different market" line from "The basics of core market mechanics," and the closing question of the last section, why a market maker would ever let a detectable imbalance sit there long enough to trade against.

Roll's Model: Why a Spread Exists Even With Zero Informed Trading

Start with the simplest possible market maker: one with no opinion about value, no information disadvantage, and no fear of anyone. They just quote a bid and ask around whatever they believe fair value is, and they get run over by order flow in both directions all day. Richard Roll asked, in 1984, whether a spread would exist even in a market like that, one with zero informed trading of any kind, and showed that yes, mechanically, it has to.

Here's the setup. Let m_t stand for the asset's true underlying value at the moment of trade t , the number you'd get if the spread didn't exist, moving as a random walk for real reasons that have nothing to do with which side of the book anyone happens to trade on. Every actual transaction, though, doesn't happen at m_t . It happens at the bid or the ask, one half-spread away from it in one direction or the other:

$$P_t = M_t + c \times q_t$$

Where:

- P_t is the transaction price actually printed at trade t .
- m_t is the true underlying value at that same moment.
- c is half the bid-ask spread, the distance from true value out to either the bid or the ask.
- q_t is the trade direction indicator: +1 if the trade happened at the ask (a buy), -1 if it happened at the bid (a sell). This is the same buyer-aggressive-versus-seller-aggressive distinction "Price, Volume, and Order Flow" already drawn out in plain language. Roll's model assumes each trade is equally likely to be either one, independent of everything before it.

That independence assumption already tells you something useful on its own. If a trade prints at the ask ($q_t = +1$) and the very next one prints at the bid ($q_{t+1} = -1$), the price just fell by $2c$, a full spread's worth, and none of that had anything to do with the asset becoming less valuable. It's what happens when transactions alternate between two fixed rails a fixed distance apart. Do that enough times in a row and consecutive price changes get a mechanical tendency to move in opposite directions.

That tendency is exactly what our autocorrelation section built the machinery to measure. Roll's move was to run that section's covariance operator, the numerator of the $p(k)$ formula, before it gets divided by variance and turned into a unitless correlation, on ΔP_t (the trade-to-trade price change) instead of on some order-flow feature, and solve backward for c . Work through the algebra, using the fact

that q_t is independent from trade to trade (the model's central, and most easily violated, assumption), and the lag-1 autocovariance of the price changes comes out to exactly negative c squared:

$$\text{Cov}(\Delta P_t, \Delta P_{t-1}) = -c^2$$

Which rearranges into an estimator you can compute from nothing but a column of trade prices:

$$\text{Spread} \approx 2 \times \sqrt{(-\text{Cov}(\Delta P_t, \Delta P_{t-1}))}$$

Where:

- $\text{Cov}(\Delta P_t, \Delta P_{t-1})$ is that same lag-1 covariance just applied to raw price changes instead of a feature. It comes out negative under this model, which is why the minus sign sits inside of the square root, you're flipping a negative number positive before taking its root.
- The 2 out front converts c (half the spread) back into the full spread.

For example, say you pull a session's worth of NQ trade prints, convert them to tick-denominated price changes, and compute the lag-1 autocovariance the same way "Autocorrelation" walked through, and you get negative 0.25 (in ticks squared). The square root of 0.25 is 0.5, times 2 is 1.0. Roll's model says the effective spread over that sample was about 1 tick. That happens to match what a one-tick-wide NQ market looks like directly on the DOM, which is a reasonable sanity check rather than a coincidence: under quiet, orderly conditions, the bounce really is close to the whole story.

Here's the limitation, and it's the important one. Roll's model assumes $q \square$ carries zero information, every trade a coin flip, uncorrelated with anything that happens to the true value afterward. It can give you an estimate of the total spread, but it cannot tell you how much of that spread is pure mechanical order-processing cost (what this model captures) versus compensation the market maker is demanding because some fraction of the people trading against them know something they don't. And in keeping with "Hypothesis Testing: p-values, Confidence Intervals, and the Cost of Testing Everything," that spread estimate deserves the same standard-error treatment before you trust it: a spread pulled from 30 trades carries exactly the same wide uncertainty band the \$50 expectancy number did back in that section. Roll's model is silent on the information question entirely. Answering it means giving up the assumption that every trade is equally uninformed, which is exactly what the next model does.

Glosten-Milgrom: The Market Maker's Fear of Knowing Less Than You

Glosten and Milgrom, in 1985, built the model Roll's leaves out: one where some fraction of the order flow actually does know something. Not a conspiracy, not "smart money" hunting anyone, the "Smart Money" Conspiracy Ecosystem section of this paper already dealt with that framing directly, just a plain admission that in any real market, some traders have better information about where fair value sits than the person quoting the bid and ask does. The market maker's problem is that they cannot tell, on any single incoming order, whether they're facing one of those traders or an ordinary participant who wants five contracts because it's Tuesday.

Set it up like this. Assume the asset's true value is one of two numbers, $V \square$ (the higher of two possible fair values) or $V \square$ (the lower one), each equally likely before any order arrives. A fraction α (alpha) of incoming orders come from informed traders, who already know which one is correct and trade accordingly, buy if the truth is $V \square$, sell if it's $V \square$, every time. The remaining $(1 - \alpha)$ of orders come

from uninformed, liquidity-driven traders, who buy or sell with equal probability regardless of the true value, for reasons that have nothing to do with information, they need to hedge something, rebalance something, or just want the exposure.

The market maker can't see which type placed any individual order, only a buy or a sell. So they set their ask equal to what the true value is expected to be given that a buy just arrived, and their bid equal to what it's expected to be given that a sell just arrived. Work through that conditional expectation with plain Bayes' rule, weighting each scenario by how likely a buy order actually is under it, and the two numbers collapse into one clean result for the spread itself:

$$Spread = \alpha \times (V_h - V_l)$$

Where:

- α (alpha) is the fraction of order flow that comes from informed traders, a number between 0 and 1.
- V_h and V_l (read: V-high and V-low) are the two possible true values the market maker believes the asset could be worth, so $(V_h - V_l)$ is just the size of that uncertainty, how far apart the two scenarios sit.

Notice what's missing from that formula: nothing about size, nothing about aggression, nothing about conviction. The spread the market maker charges is priced entirely off two things, how likely they are to be facing someone who knows more than them, and how much it would cost them if that person did.

For example, take the exact contrast already used earlier in this paper when introducing heteroskedasticity: a quiet Tuesday afternoon versus the ten minutes after a CPI print. On the quiet Tuesday, say α sits around 0.10 (one in ten orders is informed) and the market's honest uncertainty about fair value is a tight 20 ticks. $Spread = 0.10 \times 20 = 2$ ticks. Ten minutes after CPI, both numbers move against the market maker at once: α might jump to 0.30 (informed participants are far more active immediately after new information lands) while the uncertainty band about fair value widens to 30 ticks (nobody, informed or not, is entirely sure yet where the number settles). $Spread = 0.30 \times 30 = 9$ ticks. Spreads don't blow out around news releases because market makers panic. They blow out because α and $(V_h - V_l)$ both went up, and the formula says the toll goes up with them, which is the same widening "Spread and execution cost" already told you to price into every trade, just without a mechanism attached yet.

Two honest caveats, in keeping with how this paper has treated every other formula. First, this is the single-round version. The real Glosten-Milgrom setup runs this update trade by trade across a whole sequence of orders, with the market maker's beliefs about V_h versus V_l converging as more flow arrives, which is a large part of why a spread that blows out on a surprise tends to compress again over the following minutes rather than staying wide forever: the market maker is learning in real time, the same way anyone updates on new evidence. Second, and this is the point worth sitting with: nowhere in this model's construction do informed traders need to be identifiable, malicious, or coordinated. It only requires α to be greater than zero. That's the actual, formal answer to "insider trading drives everything" and "whales are hunting your stops": informed trading is real, measurable, and alpha-sized, and it still has nothing to do with anyone hunting anyone. The market maker isn't defending against a person. They're defending against a probability, and they'd charge the same toll to a room full of strangers they'll never identify.

Kyle's Lambda: Price Impact Has a Slope

"The basics of core market mechanics" gave you one sentence on this and moved on: "A trade's effect is not determined by size alone. The same 100-lot can be irrelevant in deep liquidity and meaningful in a thin, fast, or news-driven book." Albert Kyle, in 1985, is the model that turns that sentence into a number. It's also, not coincidentally, the thing sitting behind the one term in the all-in-cost formula from "Spread and execution cost" that got named and never explained: commissions plus exchange fees plus spread cost plus slippage plus market impact. That last term is what this section is actually about.

Kyle's setup adds one piece to Glosten-Milgrom: instead of only asking how the market maker should price an order they can't identify, it asks how an informed trader who does know the true value should behave, given that they know the market maker is watching total order flow and pricing off it. The informed trader's best move is never to dump their whole position in one print, that would make the imbalance obvious and get priced against them immediately. The equilibrium has them spread their trading out, sized just large enough to profit from what they know, small enough to stay camouflaged inside whatever noise trading is happening anyway. The market maker, for their part, can't isolate the informed trader's share of that flow either. All they observe is the total, and all they can do is set a price that moves in proportion to it, protecting themselves on average across every informed and uninformed order combined.

That proportionality constant is Kyle's Lambda, and in practice it's what you'd actually estimate by regression:

$$\Delta P_t = \lambda \times Q_t + \varepsilon_t$$

Where:

- ΔP_t is the price change over some short interval, say one minute of NQ trading.
- Q_t is the net signed order flow over that same interval (buy volume minus sell volume, in contracts), built from the same kind of aggressor classification "Trades and Prints" already warned isn't always clean.
- λ (lambda) is the price impact per unit of order flow, the slope, in ticks or points per contract. This is the market impact term from "Spread and Execution cost," now with a formula behind it.
- ε_t (epsilon, read: the error at time t) is everything about the price change that isn't explained by order flow at all, new information hitting every participant at once, unrelated to anyone's individual trade.

Kyle's model also tells you what λ is built out of, not just what it looks like in a regression:

$$\lambda = \sigma_v / (2 \times \sigma_u)$$

Where:

- σ_v (read: sigma-v) is the standard deviation of the market's honest uncertainty about the asset's true value, the same kind of quantity as $(V_h - V_l)$ above, just expressed as a spread of outcomes instead of two discrete scenarios.
- σ_u (read: sigma-u) is the standard deviation of noise, uninformed order flow, how much liquidity-driven trading is happening that has nothing to do with information.

That ratio is a direct, quantified version of everything "Trading sessions and liquidity" already told you qualitatively: more noise trading (bigger σ_u) gives an informed trader more crowd to hide in, so λ goes down and the same size order moves price less, a deeper market. More fundamental uncertainty (bigger σ_v) means any given order is more likely informed relative to the noise around it, so λ goes up and the same size order moves price more. Nothing about either half of that ratio depends on anyone's identity. It's two variances, divided.

For example, say you regress one-minute NQ price changes (in ticks) on one-minute net signed volume (in contracts) during regular trading hours and get an estimated λ of 0.002 ticks per contract. A 500-contract net imbalance in a minute would be expected to move price about $500 \times 0.002 = 1$ tick. Run the same regression on the same contract overnight, when σ_u drops off a cliff and displayed depth thins out along with it, and λ might come out closer to 0.01. That same 500-contract imbalance now moves price about $500 \times 0.01 = 5$ ticks, five times the impact, for the exact same order flow, purely because there's less noise around it to absorb it into.

This is also the direct, formal answer to the question this paper closed on before this section started: why would a market maker ever let a detectable imbalance sit there long enough to trade against? Because in Kyle's equilibrium, the informed trader is actively working to make sure it never looks detectable in the moment. λ is priced in advance, as an average cost across all order flow, informed and noise combined. The market maker isn't failing to notice one large, obvious, informed trade. There usually isn't one. There's a long sequence of ordinary-looking trades, deliberately sized to blend in, and λ is the toll charged on all of them because a few, unidentifiable in real time, are informed. That's the same warning "What Data Cannot Reveal" already gave you, now with an actual number attached to it: a print proves execution, not motive, and Kyle's model is the formal reason no amount of staring at one print will ever recover the motive after the fact.

One Order Book, Three Tolls

Model	What it explains	Core assumption	What it leaves out
Roll (1984)	Why a spread exists mechanically, with zero information	Trade direction is a coin flip, independent over time	independent over time Can't separate order-processing cost from adverse-selection cost
Glosten-Milgrom (1985)	Why the spread widens specifically with information risk	A fixed fraction, α , of order flow is informed	Static single round; says nothing about how imbalance itself moves price
Kyle (1985)	Why order flow imbalance moves price, and by how much	An informed trader trades strategically to hide inside noise	Doesn't touch the mechanical bounce Roll already explained; needs $Q^\square$ built from real, imperfect trade classification

Put together, these three models cover the whole gap between "why does a bid and ask exist at all" and "why did NQ just move five ticks on a 500-lot imbalance." Roll explains the floor: a spread that would exist even with zero informed trading whatsoever, just the mechanics of alternating between two rails. Glosten-Milgrom explains why that floor rises: a market maker who can never identify informed traders individually still has to charge everyone more as α , the probability any given order carries information, goes up. Kyle explains the consequence once that pricing is in place: a linear, measurable relationship between signed order flow and price, with a slope that depends on how much noise trading exists to hide inside.

None of that is a number you can pull off the DOM in real time yet. Roll's spread estimator, Glosten-Milgrom's α , and Kyle's λ are parameters of someone else's stylized model of trading, built to be provably correct under a specific set of assumptions, not features sitting in a data feed. Turning each of those ideas into something you can actually compute, tick by tick, off real order book and trade data, and being honest about exactly what gets lost translating theory into a live number, is where this paper goes next.

Roll's spread estimator, Glosten-Milgrom's α , and Kyle's λ are, as the last section closed on, parameters of someone else's stylized model, not something sitting in your NQ data feed. This section is the promised translation, built in the order these things actually depend on each other: a single corrected price first, then a snapshot measure of imbalance, then a properly dynamic measure of order flow, then a full attempt at estimating something like Glosten-Milgrom's α directly. Each one is a strict prerequisite for the next.

Microprice: The Order Book's Best Guess at Fair Value

Roll's model needed m , the asset's true underlying value, the number you'd get if the spread didn't exist. Roll's model treats m as an unobservable abstraction sitting underneath the bid and ask. Microprice is the practical, computable attempt to estimate that same number using nothing but what's sitting on the DOM right now: the best bid, the best ask, and the size resting at each.

The naive candidate is the midpoint, $(bid + ask) / 2$, which "The Order Book" already gave you the ingredients for. But the midpoint treats a thick bid and a thin ask exactly the same as a thin bid and a thick ask, and that throws away real information. Think of the two queues as racing each other: every incoming market order eats into either the resting bid size or the resting ask size, and under a simple assumption that buy and sell market orders arrive equally often (Roll's own null assumption, no informed trading, just a coin flip), whichever queue is thinner is more likely to run out first. If the ask queue empties first, that ask level disappears and price ticks up. If the bid queue empties first, price ticks down. A small gambler's-ruin argument on that race gives the probability of an up-tick as:

$$\frac{Q_{bid}}{(Q_{bid} + Q_{ask})}$$

With heavier size sitting at the bid making the thinner ask more likely to give way first.

Weight the bid and ask prices by those probabilities and you get microprice:

$$Microprice = \frac{(Q_{bid} \times P_{ask} + Q_{ask} \times P_{bid})}{(Q_{bid} + Q_{ask})}$$

Where:

- P_{bid} and P_{ask} are the current best bid and ask prices
- Q_{bid} and Q_{ask} are the displayed resting size at those two prices.
- Notice the crossed weighting: P_{ask} gets weighted by Q_{bid} and P_{bid} gets weighted by Q_{ask} . A heavy bid pulls the estimate up toward the ask, a heavy ask pulls it down toward the bid, exactly the race described above.

For example, say NQ is quoted 21000.00 bid, 21000.25 ask, with 300 contracts resting at the bid and 100 at the ask. The naive midpoint is 21000.125. Microprice = $(300 \times 21000.25 + 100 \times 21000.00) / 400 = (6,300,075 + 2,100,000) / 400 = 21000.1875$, noticeably closer to the ask than the naive midpoint, because the heavier bid size makes an up-tick the more likely next move. Flip the sizes, 100 at the bid and 300 at the ask, and microprice drops to 21000.0625, closer to the bid instead.

Microprice is still just a top-of-book estimate. It says nothing about the nine price levels sitting behind the best bid and ask, and it inherits every caveat "Queue Order and Priority Fills" already raised: the size behind that weighting can be canceled, refreshed, or partially filled the instant after you compute it. It's a better m than the naive midpoint, not a perfect one.

Order Book Imbalance

The ratio sitting inside the microprice formula is useful enough on its own that it gets used as a standalone feature, stripped of the price weighting entirely:

$$OBI = \frac{(Q_{bid} - Q_{ask})}{(Q_{bid} + Q_{ask})}$$

Where:

- Q_{bid} and Q_{ask} are the same best-bid and best-ask resting sizes as above
- OBI is bounded between -1 (nothing resting at the bid, all size at the ask) and +1 (the reverse), with 0 meaning the book is balanced.

This is exactly one of the three features named back in the discussion of stationarity and spurious regression, orderflow imbalance, order book imbalance, and microprice deviation from mid price, as things built to oscillate around a fixed reference point rather than wander the way raw price does. That's precisely the property that makes OBI usable as a research feature at all, and it's worth confirming with an actual stationarity test on your own sample rather than assuming it, per that same section.

For example, using the numbers above, $OBI = (300 - 100) / 400 = 0.5$, a fairly strong buy-side imbalance. Flip the sizes and $OBI = (100 - 300) / 400 = -0.5$.

OBI's weakness is that it's a single snapshot of exactly the kind of data "Queue Order and Priority Fills" already warned about: displayed liquidity that isn't a durable commitment. A resting order feeding into Q_{bid} can be pulled the instant after you sample it, and nothing stops a participant from resting size specifically to shift OBI's reading and then canceling before it ever has to trade, a real, documented practice generally referred to as spoofing. OBI also only looks at the best price level, ignoring the other nine visible in a full MBP feed. None of this makes it useless, it means treating a single OBI reading as a signal rather than as one noisy input is exactly the kind of overconfidence "Overfitting and Data Mining" already told you to distrust. What OBI is missing is any sense of flow, whether the imbalance is building over time or is a one-off snapshot about to reverse, which is what the next metric adds.

Order Flow Imbalance: A Cleaner Version of Kyle's Q

Kyle's lambda needed $Q_{\square}$, the net signed order flow over some interval, and building that in practice meant classifying every trade as buyer-initiated or seller-initiated, exactly the aggressor classification "Trades and Prints" already flagged as imperfect without richer data. Order flow imbalance, developed by Cont, Kukanov, and Stoikov in 2014, sidesteps that problem entirely by never classifying a single trade. It's built only from changes to the best bid and ask, price and size, which are directly observable without inferring anyone's intent.

Every time the top of book updates, score that single update with one signed number: positive if it looks bullish, more buying pressure or less selling resistance, negative if it looks bearish.

On the bid side:

- Bid price rises: contribution = +(the new resting size). A better bid appeared that wasn't there a moment ago.

- Bid price unchanged: contribution = +(the change in size). More contracts joining the bid queue is positive, contracts leaving is negative.
- Bid price falls: contribution = -(the previous resting size). That whole level of buying support just vanished.

On the ask side, the same three rules, but read in reverse, since a stronger ask means more selling pressure, not more buying pressure:

- Ask price falls: contribution = -(the new resting size). The ask just got more aggressive, more willing sellers at a better price.
- Ask price unchanged: contribution = -(the change in size). More contracts joining the ask queue is negative, contracts leaving is positive.
- Ask price rises: contribution = +(the previous resting size). That level of selling resistance just vanished.

Add up every contribution, bid-side and ask-side alike, across every book update inside interval t : $OFI_t = \text{the sum of all signed contributions over interval } t$. Then run the same regression kyle's section already introduced, swapping in this new regressor:

$$\Delta P_t \approx \lambda' \times OFI_t + \varepsilon_t$$

Where:

- OFI_t is the quantity just defined.
- λ' (lambda-prime) is a price-impact coefficient playing the same role as Kyle's λ , but it isn't the same number, since OFI_t and raw signed trade volume aren't the same regressor.
- ε_t is, same as before, whatever part of the price change isn't explained by order flow.

For example, over one second of NQ trading, three book updates happen: the bid ticks up from 21000.00 to 21000.25 with 40 new contracts resting there (bid price rose, contribution +40), ask size at 21000.50 grows from 60 to 90 with no price change (ask price unchanged, contribution $-(90 - 60) = -30$), and the bid size at 21000.25 then shrinks from 40 to 25 with no price change (bid price unchanged, contribution $+(25 - 40) = -15$). $OFI = 40 - 30 - 15 = -5$. Notice the bid ticked up, which sounds bullish read off a chart, but the net OFI over the same window is slightly negative once the cancellation and the growing ask are accounted for. That disagreement is exactly the kind of information a candle, or even a naive tick-direction count, throws away.

Cont, Kukanov, and Stoikov found this regression fits short-horizon price changes considerably better than the same regression run on raw signed trade volume, precisely because OFI captures cancellations and new resting orders that never executed at all, order flow that Kyle's original formulation, built purely from executed trades, simply can't see.

VPIN: Estimating Alpha in Volume Time, and the Fight over Whether It Works

Glosten-Milgrom's α , the fraction of order flow that's informed, is a parameter of a stylized model, not something sitting in a data feed. Easley, López de Prado, and O'Hara built VPIN (Volume-Synchronized Probability of Informed Trading) in 2012 as an attempt to estimate something like it directly from trade data, and the first decision it makes is to throw out calendar time entirely.

Recall the heteroskedasticity example already used twice in this paper: a quiet Tuesday afternoon and the ten minutes after a CPI print aren't drawn from the same distribution. Trading intensity itself is nonstationary across calendar time, a one-minute bar during the cash open and a one-minute bar at 3 a.m. overnight can hold wildly different amounts of information, purely because they hold wildly different amounts of volume. VPIN's fix is to stop measuring in minutes and start measuring in contracts: divide trading into buckets that each contain a fixed volume, say 1,000 contracts, however long or short a stretch of calendar time it takes to fill one. A busy bucket might fill in ten seconds, a quiet one might take twenty minutes, but every bucket represents the same amount of trading activity by construction.

Within each bucket, VPIN needs a buy volume and a sell volume, and rather than classifying individual trades, the same aggressor problem OFI sidestepped by avoiding entirely, VPIN sidesteps it a different way: Bulk Volume Classification looks at the bucket's overall price change and asks how much of the bucket's volume was probably buy-initiated, using the standard normal distribution:

$$V_{buy} = V \times \Phi(\Delta P / \sigma)$$

Where:

- V is the fixed bucket size, the same 1,000 contracts in every bucket.
- ΔP is the price change from the start of the bucket to the end of it.
- σ is the standard deviation of that bucket-level price change, estimated from recent history.
- Φ is the standard normal cumulative distribution function, the exact same machinery "Hypothesis Testing: p-values, Confidence Intervals, and the Cost of Testing Everything" already introduced for turning a standardized score into a probability, repurposed here to turn a standardized price move into a buy/sell split instead of a p-value.
- $V_{sell} = V - V_{buy}$

A bucket where price rose a lot relative to its usual bucket-to-bucket noise gets classified as mostly buy volume. A bucket that barely moved gets split close to even. VPIN itself is just the average absolute imbalance across the most recent n buckets:

$$VPIN = \frac{(\text{sum of } |V_{buy} - V_{sell}| \text{ across the last } n \text{ buckets})}{(n \times V)}$$

For example, with a bucket size of 1,000 and σ of 2 ticks: a bucket where price rose 3 ticks gives a standardized move of 1.5, and $\Phi(1.5)$ is about 0.93, so $V_{buy} = 933$, $V_{sell} = 67$, and $|V_{buy} - V_{sell}| = 866$, or 0.966 of that bucket's volume. A calmer bucket where price only moved 0.2 ticks gives a standardized move of 0.1, $\Phi(0.1)$ is about 0.54, so the split is close to even and the imbalance is only about 0.08 of the

bucket. Average enough buckets of both kinds together and you might land on a VPIN reading of, say, 0.30.

Here's the real academic fight, and it belongs in this paper for the same reason "Overfitting and Data Mining" and "Hypothesis Testing" exist at all: a metric having real economists' names attached to it doesn't exempt it from scrutiny. Easley, López de Prado, and O'Hara originally presented VPIN as having risen sharply ahead of the May 2010 Flash Crash, framing it as an early-warning measure of toxic liquidity. Andersen and Bondarenko, in a direct response published in 2014, argued this doesn't hold up well under closer inspection. Their core objections: VPIN's construction makes it mechanically correlated with volatility and trade size almost by definition, since a bucket with a large price move gets classified as heavily imbalanced by the BVC formula regardless of whether any actual informed trading caused that move, so a rise in VPIN can just as easily be picking up ordinary volatility clustering as anything resembling Glosten-Milgrom's α . They also showed VPIN's apparent predictive power is sensitive to the exact bucket size and averaging window chosen, the same researcher-degrees-of-freedom problem "Overfitting and Data Mining" already told you to distrust in any backtest. Whether VPIN adds real information beyond a simpler, transparent volatility measure remains a genuinely contested empirical question in the literature, not a settled one, and the honest position is the same one this paper opened with: unproven until it survives a fair baseline and out-of-sample testing, no matter whose name is attached to it.

Four Numbers, Same Rules

Metric	What it estimates	Built from	Key limitation
Microprice	A corrected fair-value price, a practical estimate of Roll's m	Best bid/ask price and size	Still just top-of-book; ignores everything resting deeper in the book
Order Book Imbalance	A snapshot of resting buy vs. sell pressure	Best bid/ask size only	A snapshot of displayed liquidity that can be canceled or spoofed before it's ever tested
Order Flow Imbalance	Kyle's Q , rebuilt from quote and cancellation events instead of classified trades	Every book update at the best bid/ask	Needs a full, correctly time-stamped event feed; noisy at very short intervals
VPIN	Glosten-Milgrom's α , estimated in volume time instead of calendar time	Volume-bucketed, BVC-classified buy/sell splits	Andersen and Bondarenko's critique: much of its apparent predictive power may be mechanical, driven by volatility and bucket construction rather than genuine informed-trading detection

None of these four numbers is a trading rule. Microprice is a better m□ than the naive midpoint. OBI is that same information stripped down to a bounded snapshot. OFI is Kyle's Q rebuilt from quote events instead of classified trades. VPIN is an attempt at Glosten-Milgrom's α, measured in volume time, and it's also the clearest proof in this paper that a real academic pedigree doesn't buy a metric a pass from "Baselines and Comparisons" or "Robustness Testing." Every one of these numbers still has to earn its place the same way a moving-average crossover would.

What none of them do on their own is tell you what to do next. A microprice, an OBI reading, an OFI□, and a VPIN reading are each just a number at a point in time, continuous, noisy, and one observation doesn't tell you what regime you're actually in. Turning a feature like that into something you can act on means collapsing it into a small number of states, quiet versus imbalanced versus toxic, and asking a very specific question: does the market's next state depend only on its current one, or is it dragging around more history than that. That's a modeling question, not a definition question, and it's where this paper goes next.

The closing question of the last section was specific on purpose: does the market's next state depend only on its current one, or is it dragging around more history than that. Answering that needs two things this section builds in order: an actual state space to work with, since none of microprice, OBI, OFI□, or VPIN come pre-cut into states, and then a formal test of the question itself, rather than an assumption borrowed from wherever you first read the word "Markov."

Discretizing a feature: From a Number to a State

Take OFI□, the feature built in the last section, and cut it into a small number of buckets using its own empirical distribution rather than any round number that happens to look clean on a chart. A common, defensible choice is terciles: compute the 33rd and 67th percentile of OFI□ over your in-sample period, then label every observation below the lower cutoff as Down-pressure (D), everything above the upper cutoff as Up-pressure (U), and everything between as Neutral (N). Those two cutoffs get fixed once, on the in-sample period only, and then applied unchanged to any out-of-sample data, the same discipline "In-Sample and Out-of-Sample Testing" already demanded of any other parameter choice. Letting the boundaries recompute themselves on new data as it arrives is a second, quieter way to leak the future into your model.

How many states to use is its own tradeoff, and it's the same one the data granularity ladder at the start of this paper already walked through, just one level removed. More states give you a finer, more descriptive picture of what "up-pressure" actually means, five gradations of imbalance instead of three. But every additional state multiplies the number of transition probabilities you need to estimate reliably, and "Overfitting and Data Mining" already told you what happens to a model with too many adjustable pieces and not enough data to support them. Three states, kept, for the rest of this section.

The Transition Matrix: What Happens Next, Conditioned on Now

With X□ now standing for the discretized state at time t, one of D, N, or U, the Markov property is a specific, testable claim about how X□ evolves:

$$P(X_{t+1} = j | X_t = i, X_{t-1} = i', \dots, X_0 = i_0) = P(X_{t+1} = j | X_t = i)$$

Where:

- X_t is the discretized state after time t .
- $i, i', \dots, i_0$ and j are just placeholder labels for whichever specific states the process happened to visit; the formula says the same thing regardless of which ones they are.

In plain language: knowing the entire history of states up to now adds nothing beyond just knowing the current one. Whatever information X_{t-1} and everything before it carried is already fully absorbed into X_t . This is a strong claim, and it's exactly the kind of claim this paper has spent its whole statistical toolkit building the habit of testing rather than assuming. If it holds, the whole process is described by one small object, the transition matrix, with one row per current state and one column per next state:

$$P_{ij} = P(X_{t+1} = j \mid X_t = i)$$

Estimated directly from data by counting:

$$P_{ij} = N_{ij} / N_i \cdot$$

Where:

- N_{ij} is the number of times state i was immediately followed by state j , somewhere in your sample.
- $N_i \cdot$ (read: N sub i , dot) is the total number of times state i occurred as a starting point, summed across everywhere it went next. The dot just means "add over every possible destination."
- Every row of P sums to 1 by construction, since from any state i , the process has to go somewhere.

For example, say a sample of discretized OFI gives these transition counts:

From \ To	D	N	U	Total
D	40	50	10	100
N	30	40	30	100
U	10	50	40	100

Dividing each row by its total gives the transition matrix:

D: [0.40, 0.50, 0.10] N: [0.30, 0.40, 0.30] U: [0.10, 0.50, 0.40]

Each row sums to 1.00, confirm that before trusting anything downstream of it. Read row D: from Down-pressure, there's a 40% chance of staying D, a 50% chance of drifting back to N, and only a 10% chance of flipping straight to U. Row U mirrors it exactly. Row N is the most balanced: a 40% chance of staying N, and an even 30/30 split toward either D or U, no directional lean from the neutral state.

This one small matrix also answers a question one step beyond "what happens next": what happens in two steps. The Chapman-Kolmogorov equation says the two-step transition probabilities are just the matrix multiplied by itself, P^2 , so starting from D, the probability of landing in each state two steps later is:

$$P^2(D, j) = \sum_k P(D, k) \times P(k, j)$$

For example, the probability of being back at D two steps after starting at D:

$$P^2(D, D) = (0.40 \times 0.40) + (0.50 \times 0.30) + (0.10 \times 0.10) = 0.16 + 0.15 + 0.01 = 0.32$$

$$P^2(D, N) = (0.40 \times 0.50) + (0.50 \times 0.40) + (0.10 \times 0.50) = 0.20 + 0.20 + 0.05 = 0.45$$

$$P^2(D, U) = (0.40 \times 0.10) + (0.50 \times 0.30) + (0.10 \times 0.40) = 0.04 + 0.15 + 0.04 = 0.23$$

Check: $0.32 + 0.45 + 0.23 = 1.00$. Starting from D, the one-step forecast said 40% D / 50% N / 10% U. Two steps out, that's already shifted to 32% D / 45% N / 23% U, the probability of reaching U more than doubled. That's the entire payoff of the Markov property holding: every multi-step forecast, arbitrarily far out, comes from nothing but powers of this one small matrix. If the property doesn't hold, P^2 computed this way is simply wrong, since it silently assumes X_{t-1} told you nothing that X_t didn't already capture.

Testing the Markov Property: Does One Step of Memory Cover Everything?

Everything in the last section assumed the Markov property first, then built on top of it. Testing it means comparing that assumption directly against a richer, second-order model, one where the transition from j to k is allowed to depend on the state before j too, and checking whether the extra dependence is actually doing any work. The null hypothesis is exactly the claim the last section made on faith:

$$H_0: P(X_{t+1} = k \mid X_t = j, X_{t-1} = i) = P(X_{t+1} = k \mid X_t = j), \text{ for every } i, j, k$$

If H_0 is true, knowing two steps of history (i, j) predicts the next state no better than knowing one step (j) alone, so the expected count of any three-state sequence (i, j, k) should just be the two-step history count times the plain first-order transition probability:

$$E_{ijk} = N_{ij} \times P(k \mid j) = N_{ij} \times (N_{jk} / N_{j\cdot})$$

Compare that expectation against what actually happened, N_{ijk} , the observed count of the exact sequence i then j then k , with a chi-squared statistic:

$$\chi^2 = \sum_i \sum_j \sum_k (N_{ijk} - E_{ijk})^2 / E_{ijk}$$

This is the same kind of joint significance question the Ljung-Box test already asked of a continuous feature back in "Autocorrelation: Does the Past Actually Predict the Future?", just rebuilt for a categorical state sequence instead of a numeric one.

The degrees of freedom come from counting parameters directly, the same nested-model logic behind every chi-squared test in this paper. An m -state first-order model needs m rows, each with $(m - 1)$ free probabilities (the last one is forced by the row summing to 1), for $m(m - 1)$ total. A second-order model conditions on m^2 possible (i, j) pairs instead of just m states, each again with $(m - 1)$ free probabilities, for $m^2(m - 1)$ total. The difference between them is the degrees of freedom:

$$df = m^2(m - 1) - m(m - 1) = m(m - 1)^2$$

For the three-state D/N/U chain, $df = 3 \times (3 - 1)^2 = 12$.

To keep the arithmetic checkable by hand, shrink to two states for a moment, Up (U) and Down (D), where $m = 2$ gives $df = 2 \times 1^2 = 2$. Say a sample of two-step histories gives these counts:

History (i, j)	→ U	→ D	Total ($N_{i\Box}$)
U, U	30	20	50
D, U	24	16	40
U, D	15	35	50
D, D	12	28	40

Pool across i to get the plain first-order rates: from state U, 54 of 90 pooled transitions land on U, so $\hat{P}(U|U) = 0.60$, $\hat{P}(D|U) = 0.40$. From state D, 27 of 90 land on U, $\hat{P}(U|D) = 0.30$, $\hat{P}(D|D) = 0.70$. Multiply each $N_{i\Box}$ by those pooled rates to get the expected counts under H_0 , and compare:

History (i,j)	→ k	Observed	Expected	Contribution
U, U	U	30	$50 \times 0.60 = 30$	0
U, U	D	20	$50 \times 0.40 = 20$	0
D, U	U	24	$40 \times 0.60 = 24$	0
D, U	D	16	$40 \times 0.40 = 16$	0
U, D	U	15	$50 \times 0.30 = 15$	0
U, D	D	35	$50 \times 0.70 = 35$	0
D, D	U	12	$40 \times 0.30 = 12$	0
D, D	D	28	$40 \times 0.70 = 28$	0

$\chi^2 = 0$. These particular counts were built so the ratio of U to D stayed exactly 60/40 whether the history was (U,U) or (D,U), and exactly 30/70 whether it was (U,D) or (D,D), on purpose, to make the arithmetic land on a clean number you can check by hand. Real data will essentially never hand you a

literal zero. What you're actually checking χ^2 against is a critical value from the chi-squared distribution at $df = 2$, about 5.99 at the usual 5% significance level. Zero sits nowhere close to 5.99, so this sample gives no evidence against H_0 : one step of memory looks like it's enough. Run the real three-state version on your own OFI sequence and you'd compare your computed χ^2 against roughly 21.03 instead, the $df = 12$ critical value.

One practical caveat before trusting any of this: the chi-squared approximation itself needs enough data per cell to be valid, a common rule of thumb is expected counts of at least 5 in the large majority of cells. The three-state, second-order test has 27 cells (nine (i,j) pairs times three possible k), each of which needs to be filled by actual transitions in your sample before the test means anything. That's the same lesson "Hypothesis Testing: p-values, Confidence Intervals, and the Cost of Testing Everything" already taught with plain trade counts: more parameters means each one is estimated off a thinner slice of your data, and a test built on cells with only two or three observations in them isn't testing anything, it's measuring sampling noise and calling it a rejection.

If the test does reject H_0 , the honest reading isn't "throw out the whole framework." It's that a plain first-order chain, at this state definition, isn't capturing everything relevant, and the two realistic fixes are either a genuine second-order model (at the cost of that $m^2(m-1)$ parameter count, and everything "Overfitting and Data Mining" already told you about parameter counts you can't actually support) or coarser states that absorb more of that missing structure into fewer, better-estimated categories. Either way, it's worth checking whether the transition matrix itself is even stable across the regimes "Time, Volatility, and Regime" already told you to separate, since a matrix pooled across the cash open and 3 a.m. overnight might fail this test for reasons that have nothing to do with memory and everything to do with mixing two different processes into one estimate.

The Model Is Just Another Claim to Test

Nothing about the transition matrix or the order test above is special because it has the word "Markov" attached to it. It's built the same way, and it has to survive the same scrutiny, as the expectancy formula, the autocorrelation function, and VPIN's α before it: state the model precisely, estimate it on in-sample data, and test whether it actually holds up rather than assuming it does because the algebra is convenient.

None of this, the discretization, the matrix, the order test, tells you how long any structure sitting inside that matrix survives before it decays. A transition matrix estimated from six months of data assumes the process is stable enough that the numbers built from January still describe June, and "Time, Volatility, and Regime" already told you that's exactly the kind of assumption you can't take for granted. There's a narrower version of that same question sitting inside the matrix itself, too: even if the Markov property holds and the transition probabilities are genuinely stable, how many ticks or seconds ahead does knowing today's state actually tell you anything useful about, before that edge decays back to noise. That's a question about decay, not about structure, and it's where this paper goes next.

Two Kinds of Decay

The closing question of "Testing the Markov Property: Does One Step of Memory Cover Everything?" was deliberately narrow: does today's state tell you anything about tomorrow's state, or does the chain forget faster than that. This section asks a related but different question, one this paper has quietly been assuming an answer to since "Expected Value, Not Win Rate" first put a number on a \$50-per-trade edge. How long does that edge actually survive once you've found it?

Worth separating two things that sound similar and aren't. "Autocorrelation: Does the Past Actually Predict the Future?" asked whether a feature predicts itself, whether this minute's OFI reading says anything about next minute's OFI reading. That's a statement about the feature's own internal memory. The question this section asks is different: does the feature, measured right now, predict something about forward price returns, and if so, for how long. That second question is the one that actually pays you. A feature can be wildly autocorrelated with itself and tell you nothing about where price goes next, and a feature can have almost no memory of its own past values and still carry real short-horizon information about returns. Don't conflate the two just because they use the same statistical machinery.

The Decay Curve: Correlation Against a Moving Horizon

Take any feature already built in this paper, OFI is the natural one going forward, and define its relationship to the forward return at horizon h :

$$\rho(h) = \text{Corr}(\text{feature}_t, \text{return}_{t \rightarrow t+h})$$

Where:

- feature_t is your feature's value at time t (OFI computed over whatever interval you've chosen)
- $\text{return}_{t \rightarrow t+h}$ is the price return measured starting immediately after feature_t is observed, running forward h seconds, ticks, or whatever unit you're working in.
- $\rho(h)$ reuses the same correlation notation from "Autocorrelation," bounded between -1 and 1, but it is now measuring the relationship between two different things, a feature and a future return, not a feature and a lagged copy of itself. Same tool, different pair of variables. Don't let the shared letter confuse the two.

One rule before you compute anything: $\text{return}_{t \rightarrow t+h}$ must start strictly after feature_t is fully known. If your feature calculation for time t peeks at any information from $t+1$ or later, even accidentally through how a bar or bucket boundary is drawn, you've built lookahead into the number and every $\rho(h)$ downstream of it is fiction. This is the same discipline "In-Sample and Out-of-Sample Testing" already demanded of parameter selection, applied here to the timestamp itself.

For example, say you build a decay curve for OFI on NQ, computing $\rho(h)$ at several horizons across a large sample. At $h = 10$ seconds, with roughly 5,000 effectively independent windows, you measure $\rho(10s) = 0.35$. Because a plain correlation coefficient's sampling distribution isn't symmetric the way a sample mean's is, the standard tool isn't the t-test from "Hypothesis Testing" directly, it's the Fisher transformation, which converts ρ into something that does behave like a normal variable:

$$z = 0.5 \times \ln[(1 + \rho) / (1 - \rho)]$$

$$SE_z = 1 / \sqrt{(n - 3)}$$

Where:

- z is the Fisher-transformed correlation, not a p-value or a t-statistic on its own, just a rescaled version of ρ that's better behaved for building a confidence interval.
- ρ is the measured correlation at that horizon.

- n is the number of effectively independent observations behind that ρ .
- SE_z is the standard error of z , playing the same role SE played back in "Hypothesis Testing," just computed differently because z isn't a sample mean.

Plug in $\rho = 0.35$, $n = 5,000$: $z = 0.5 \times \ln(1.35 / 0.65) = 0.365$, and $SE_z = 1/\sqrt{4,997} = 0.0141$. A 95% interval in z -space is $0.365 \pm 1.96 \times 0.0141$, or $[0.338, 0.393]$. Converting that back into correlation units gives a 95% confidence interval of roughly $[0.33, 0.37]$. Tight, clearly nonzero, exactly what you'd want a real short-horizon signal to look like.

Now run the same calculation at $h = 180$ seconds (three minutes). Say the measured correlation has fallen to $\rho(180s) = 0.05$. Two things change here, not just the correlation itself: 180-second forward windows overlap heavily if you're sampling every few seconds, which means your raw observation count massively overstates how much independent information you actually have. That's the same clustering caveat "Hypothesis Testing" already flagged, applied to overlapping horizons instead of clustered trades. After correcting for that overlap, say your effective sample size has dropped to $n = 800$. Same formula: $z = 0.5 \times \ln(1.05/0.95) = 0.050$, $SE_z = 1/\sqrt{797} = 0.0354$. The 95% interval is 0.050 ± 0.069 , or $[-0.019, 0.119]$ in z -space, which converts back to roughly $[-0.02, 0.12]$ in correlation units.

That interval includes zero. At 180 seconds, the honest statement is that OFI's relationship to forward returns is no longer distinguishable from noise, not because the number dropped to exactly zero, but because you don't have enough independent evidence left to claim it isn't zero. Compare the two horizons side by side: at 10 seconds you have a tight interval that excludes zero by a wide margin, at 180 seconds you have a wide interval straddling zero. That's what decay looks like when you measure it properly instead of eyeballing a chart.

Multiple Horizons, Multiple Tests

If you compute $\rho(h)$ at every horizon from 1 second out to 300 seconds and then report the single best-looking one, you've recreated the exact problem "Hypothesis Testing" already warned about under testing many rules at once. Scan 300 horizons at a 5% significance threshold and simple arithmetic says you should expect roughly 15 of them to look "significant" from sampling noise alone, even if the true relationship were flat everywhere. The fix is the same one already given: either hold each individual horizon to a much stricter threshold in proportion to how many you tested, or better, don't test horizons one at a time at all. Fit the whole curve as a single object and look at its overall shape, a real decaying signal produces a smooth, monotonically shrinking curve with tight early confidence bands, a spurious one produces a jagged curve with a handful of noisy spikes and wide bands throughout. The shape is the evidence. A single cherry-picked point on the curve is not.

Half-Life: Collapsing a Curve Into One Number

A full decay curve is the honest artifact, but it's useful to have one summary number, the same way "Measuring a Strategy" reduced a whole equity curve down to expectancy, drawdown, and a profit factor rather than making you stare at the raw curve every time. If a decay curve falls off roughly exponentially, which short-horizon order-flow signals often do, you can summarize it with a single time constant:

$$\tau = (h_2 - h_1) / \ln[\rho(h_1) / \rho(h_2)]$$

$$\text{Half-life} = \tau \times \ln(2)$$

Where;

- h_1 and h_2 are two horizons where you've measured the correlation.
- $\rho(h_1)$ and $\rho(h_2)$ are the correlations measured at those two horizons.
- τ (tau) is the decay time constant, in the same units as h . Larger τ means slower decay.
- Half-life is the horizon at which the correlation has fallen to exactly half its starting value, the same concept as a radioactive half-life, just applied to a statistical relationship instead of a physical one.

Using the two horizons already measured, $h_1 = 10s$ with $\rho = 0.35$, and adding a middle point at $h_2 = 60s$ with $\rho(60s) = 0.18$: $\tau = (60 - 10) / \ln(0.35/0.18) = 50 / \ln(1.944) = 50 / 0.665 = 75.2$ seconds. Half-life = $75.2 \times 0.693 =$ about 52 seconds. In plain terms, whatever edge OFI carries at the 10-second mark is down to roughly half its strength by the time you're 52 seconds out, a quarter of its strength by around 104 seconds, and by 180 to 210 seconds it's small enough to disappear into the confidence interval you just computed above. Check the fit against the 180-second point measured earlier: the exponential model predicts $\rho(180s) \approx 0.35 \times e^{(-170/75.2)} \approx 0.037$, reasonably close to the observed 0.05. Not a perfect match, real decay curves wobble and rarely fit a clean exponential exactly, but close enough to confirm the summary number is in the right neighborhood rather than fiction.

Two honest caveats belong here. First, τ and half-life are a compression of the full curve, not a replacement for it, the same relationship expectancy has to the full trade distribution back in "Expected Value, Not Win Rate." A single half-life number can hide a curve that isn't actually monotonic, some order-flow features show a secondary bump from mean-reversion a few seconds after the initial decay, and a single exponential fit will paper right over that bump. Second, with more than two horizons available, don't just pick a convenient pair, fit τ properly with a regression of $\ln \rho(h)$ on h across every horizon you measured, the same log-linear regression style already used to estimate Kyle's lambda and the OFI price-impact coefficient. The slope of that regression is $-1/\tau$. Two points give you an estimate, a full regression across many points gives you a defensible one.

Decay in a Transition Matrix: How Fast the Chain Forgets

The decay curve above treats the feature as continuous. The Markov framework from the last section treats it as a small number of discrete states, and decay shows up there too, just in a different form: convergence to the stationary distribution.

Take the exact D/N/U transition matrix already built in "The Transition Matrix: What Happens Next, Conditioned on Now":

D: [0.40, 0.50, 0.10]
 N: [0.30, 0.40, 0.30]
 U: [0.10, 0.50, 0.40]

Every Markov chain of this type (irreducible and aperiodic, meaning it's possible to eventually reach any state from any other and there's no rigid cycle forcing it) converges as you take more and more steps forward to a single stationary distribution, a fixed vector π (pi, and no, unrelated to the 3.14159 you're thinking of, just the standard linear-algebra letter for this object) satisfying:

$$\pi P = \pi, \text{ with the components of } \pi \text{ summing to } 1$$

Solve that system for the matrix above and you get $\pi = [0.273, 0.455, 0.273]$ for $[D, N, U]$. You can check it directly: multiply π by P and you get π back unchanged, $0.40 \times 0.273 + 0.30 \times 0.455 + 0.10 \times 0.273 = 0.273$, and the other two components check out the same way. This is the state model's version of "the market's long-run, unconditional frequency of being in each state," with no reference to what state it's in right now.

Here's where decay enters. "The Transition Matrix" section already computed the two-step forecast starting from D : $P^2(D, \cdot) = [0.32, 0.45, 0.23]$. Compare that against the one-step row, $[0.40, 0.50, 0.10]$, and against the stationary distribution, $[0.273, 0.455, 0.273]$. The one-step row is far from stationary, meaning knowing you're currently in D tells you a lot, your near-term forecast looks very different from the long-run average. The two-step row has already moved most of the way toward stationary, the gap between $P^2(D, D)$ and π_D has shrunk from 0.127 (one step) to 0.047 (two steps), a better than two-thirds reduction in a single additional step. Once a row's distance from π has collapsed to something you can't distinguish from zero given your sample size, knowing today's state buys you nothing about that many steps ahead. That's the exact same decay concept as the correlation curve above, just measured as a distance between probability vectors instead of a correlation coefficient.

The formal driver of how fast that convergence happens is the matrix's second-largest eigenvalue in absolute value, sometimes called the spectral gap, a smaller second eigenvalue means faster forgetting. You don't need to compute eigenvalues by hand to get a working answer, though, the same hands-on check just performed, computing P , then P^2 , then P^3 and watching the row-to-stationary distance shrink, tells you empirically how many steps of memory your particular chain actually has before it's functionally random.

Decay Is a Regime, Not a Constant

Every number in this section, $\rho(h)$, τ , half-life, the speed of convergence to π , was estimated on one sample, over one stretch of time. "Time, Volatility, and Regime" already told you that a quiet Tuesday afternoon and the ten minutes after a CPI print aren't drawn from the same distribution. That applies here with extra weight, because decay itself is a regime-dependent quantity, not just the feature or the returns individually. A market with more noise trading around it (larger σ_u , in Kyle's terms) gives informed flow more room to hide and generally lets a signal persist a little longer before it gets arbitrated away or absorbed. A market with a thin book and elevated α , right after a news print, tends to both spread wider and decay faster, the same imbalance that would predict a real move on a quiet afternoon gets priced in and gone within a couple of seconds during a fast tape.

The practical consequence is that a half-life measured once, in one regime, and treated as a permanent property of the feature is exactly the kind of unearned assumption "Robustness Testing" already told you to break on purpose. Measure the decay curve separately across regular trading hours and overnight, separately around scheduled news, separately across calm and volatile stretches, the same regime splits already listed there. A signal's decay rate is not a fact about the feature in isolation, it's a fact about the feature interacting with current market conditions, and those conditions change. This is also the honest reason a real edge needs to be re-measured on a recurring basis rather than discovered once and traded forever unchanged: not because the underlying math stopped being true, but because the τ that made it tradable at one holding period six months ago is not guaranteed to be the τ that governs it today.

Why the Holding Period Is a Measurement, Not a Preference

Put the last several sections together and a specific holding period stops being a stylistic choice, a "3-minute chart just looks right to me" decision, and becomes the direct output of a calculation. You measure $\rho(h)$ across a range of horizons, you find the horizon where the confidence interval starts including zero, and that horizon is the point past which the feature you're trading has nothing left to say.

In the illustration above, that point sat somewhere between 60 and 180 seconds. A trader holding a position built on that feature for longer than that isn't managing a trade anymore, they're holding a position with no measured edge behind it and calling it discipline. The Markov version of the same statement: once your discretized state's distance from the stationary distribution has collapsed into noise, holding a directional view justified by "the state I saw a few minutes ago" is indistinguishable from holding a view justified by nothing at all.

Two Roads to the Same Number

Approach	What it measures	Signal of full decay	What it tells your practically
Forward correlation decay curve	How strongly a continuous feature co-moves with returns at horizon h	Confidence interval for $\rho(h)$ starts including zero	Sets a holding-period boundary directly in time units
Transition matrix convergence	How much a discretized state still differs from the chain's long-run average state	Row distance from the stationary distribution π collapses toward zero	Sets how many steps ahead a state-based forecast is still worth making

Both roads get you to the same practical answer from two different directions, one continuous, one discrete, and when they roughly agree, that agreement is itself a form of robustness check, the same spirit as comparing a footprint-based observation against a simpler baseline back in "Baselines and Comparisons."

Everything built so far, stationarity, autocorrelation, standard error, hypothesis testing, Roll, Glosten-Milgrom, Kyle, microprice, OBI, OFI, VPIN, the transition matrix, and now decay, has been built as separate machinery, each piece tested on its own, small, hypothetical examples. None of it has yet been run start to finish on one real hypothesis, from a first observation through in-sample estimation, out-of-sample testing, robustness checks, and a measured holding period, in the order this paper actually built the tools. That's where this paper goes next.